Geometry, fibers, and mechanical equilibrium

Notes by Zan Ahmad

The Differential Geometry of the Heart

Based on Chapter III of
Charles S. Peskin's
Mathematical Aspects of Heart Physiology
Courant Institute Lecture Notes, 1975, pp. 127–147

Prepared October 2026

How to read these notes

The central question of these notes is simple to state: how does a system of curved, tension-bearing fibers balance a pressure load, and what does that balance imply about the geometry of a heart valve or the heart wall? Answering it requires several different meanings of curvature, a way to differentiate vectors on a curved surface, and a careful passage from individual fibers to continuum stress.

The exposition develops the argument in Chapter III of Charles S. Peskin's *Mathematical Aspects of Heart Physiology* as a continuous course. Vector calculus and elementary ordinary differential equations are the starting point. Surface geometry, tensor notation, parallel transport, membrane stress, material coordinates, and the final reduction to a shape equation are introduced when they are needed. Proofs and calculations are included so that each formula can be reconstructed rather than memorized.

The source supplied for this work is the 21-page scan `CLN-Peskin1975-3.pdf`, with printed pages 127–147. The original lecture-note collection is available at

https://math.nyu.edu/~peskin/heartnotes/index.html.

This is a pedagogical companion, not a literal transcription. The mechanical conclusions belong to the idealized models stated in the text. In particular, free sliding, negligible bending resistance, a single effective fiber family, and axial symmetry are assumptions whose roles must remain visible throughout a derivation.

Chapters 1–5 build the geometry. Chapters 6–7 apply it to valves and their motion. Chapters 8–10 develop the three-dimensional wall model and its axisymmetric reduction. Chapter 11 supplies worked problems that connect the calculations. The appendices collect notation, units, and a route back to the original pages.

A useful reading habit is to ask three questions at every equation: what object is being differentiated, which variables are held fixed, and what are the units? These questions resolve most apparent mysteries in the later calculations.

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Chapter 1

Geometry and Force

1.1 Positions, vectors, and derivatives

A point in ordinary three-dimensional space is described by a position vector

$$\mathbf{x} = (x^1, x^2, x^3) \in \mathbb{R}^3.$$

A displacement is the difference between two positions. Although both are written as triples, they play different roles: a position locates something, while a displacement describes a direction and a distance. We use the Euclidean inner product

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{3} a^i b^i, \qquad |\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}}.$$

For nonzero vectors, their angle is determined by $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \alpha$. Perpendicularity means that the dot product vanishes.

If a vector depends on a scalar parameter, its derivative is obtained by differentiating its components in a *fixed Cartesian basis*. The familiar product rule remains valid:

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a}' \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{b}'. \tag{1.1}$$

In particular, a unit vector $\mathbf{a}(t)$ satisfies

$$|\mathbf{a}|^2 = 1 \quad \Longrightarrow \quad \mathbf{a} \cdot \mathbf{a}' = 0. \tag{1.2}$$

A vector of fixed length can change only by turning; its instantaneous change is perpendicular to itself. This elementary identity underlies the definitions of curvature, parallel transport, and the constancy of fiber tension.

A moving basis requires more care. If $\mathbf{a} = a^i \mathbf{e}_i(t)$, then

$$\mathbf{a}' = \sum_i \big((a^i)' \mathbf{e}_i + a^i \mathbf{e}_i' \big).$$

The second term is indispensable in cylindrical coordinates. Omitting it would erase the centripetal acceleration of a circular path.

1.2 Cross products, oriented area, and volume

The cross product $\mathbf{a}\times\mathbf{b}$ is perpendicular to both factors, has magnitude $|\mathbf{a}||\mathbf{b}|\sin\alpha$, and is oriented by the right-hand rule. Its sign depends on order:

$$\mathbf{b}\times\mathbf{a} = -\mathbf{a}\times\mathbf{b}.$$

Thus $\mathbf{a}\times\mathbf{b}$ represents an *oriented area vector*: its magnitude is the area of the parallelogram spanned by the two vectors, while its direction chooses a side of that parallelogram.

The scalar triple product

$$J = \mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \det[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] \tag{1.3}$$

is the signed volume of the corresponding parallelepiped. Cyclic permutations preserve it; interchanging two factors reverses its sign. The vectors are linearly independent exactly when $J\neq 0$.

Why do area and volume appear so early in a discussion of the heart? Pressure is force per area, whereas a continuum force density is force per volume. Cross products and Jacobians are the mechanisms that convert coordinate increments into these physical measures.

1.3 The chain rule in geometric form

Suppose $q\colon \mathbb{R}^3\to\mathbb{R}$ is a scalar field. Its gradient is characterized by

$$\frac{\mathrm{d}}{\mathrm{d}t}q(\mathbf{x}(t)) = \nabla q(\mathbf{x}(t))\cdot\mathbf{x}'(t). \tag{1.4}$$

Thus $\nabla q\cdot\mathbf{a}$ measures change in the direction $\mathbf{a}$. For a unit direction this is change per unit distance. If q is constant on a surface and $\mathbf{a}$ is tangent to that surface, then $\nabla q\cdot\mathbf{a}=0$. Wherever $\nabla q\neq 0$, the gradient is therefore normal to the level surface.

A surface is locally a map $\mathbf{x}(u,v)$. A curve on it is a composition $\mathbf{x}(u(t),v(t))$, and the chain rule gives

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{x}_u\frac{\mathrm{d}u}{\mathrm{d}t} + \mathbf{x}_v\frac{\mathrm{d}v}{\mathrm{d}t}. \tag{1.5}$$

The subscripts mean partial derivatives: $\mathbf{x}_u$ varies u while holding v fixed. Later, $\mathbf{x}_v$ in coordinates (s,v,p) will mean a derivative holding *both* s and p fixed. A coordinate symbol has meaning only together with this information.

1.4 Force, tension, stress, and pressure

A force vector has units of force. A cable tension T is a nonnegative scalar with the same units; its direction is supplied by the cable tangent. A force per unit cable length has units force/length. A membrane tension, often called a *stress resultant*, has units force/length because it has already been integrated through the membrane thickness. A three-dimensional material stress has units force/area.

These are different quantities even when informal language calls all of them "tension." For a sheet of thickness h, a typical membrane resultant N and a typical material stress σ_{mat} satisfy

$$N\sim h\sigma_{\mathrm{mat}}.$$

For a slender cable of cross-sectional area A_c,

$$T \sim A_c \sigma_{\text{cable}}.$$

The symbol $\sim$ here means a scale estimate, not an exact constitutive law.

Pressure p has units force/area. A uniform pressure difference across an oriented surface element supplies force

$$\mathrm{d}\mathbf{F} = p\mathbf{n}\,\mathrm{d}A. \tag{1.6}$$

We choose $\mathbf{n}$ toward the applied load in this formula, or equivalently allow p to carry a sign. In a three-dimensional continuum, an isotropic compressive stress is $-p\mathrm{Id}$, and its force density is $-\nabla p$. The gradient contributes an extra factor of inverse length, producing force/volume.

1.5 Local balance from balance on a small piece

At mechanical equilibrium the total force on each sufficiently small piece is zero. For a cable segment, endpoint tensions balance a distributed load. For a membrane patch, forces across its boundary balance pressure over its area. For a volume of tissue, boundary tractions balance volume forces.

The divergence theorem explains the last passage. For a vector field $\mathbf{j}$,

$$\int_{\partial V} \mathbf{j} \cdot \mathbf{n}\,\mathrm{d}A = \int_V \operatorname{div} \mathbf{j}\,\mathrm{d}V. \tag{1.7}$$

The left side is the net outward flux; the right side sums the local flux production. If the integral of a continuous scalar function is zero over every small region, that function must vanish pointwise. Otherwise it would have a fixed nonzero sign in some neighborhood and a nonzero integral there. This observation is what turns an integral balance into a differential equation.

Example 1.1 (Dimensions predict the first curvature law)**.** A curved cable with tension T supplies a transverse load per unit length. If the only local geometric scale is its radius of curvature R, dimensional consistency suggests a load of order T/R. The exact vector result will be $T\kappa\mathbf{n}_c$, where $\kappa = 1/R$ and $\mathbf{n}_c$ points toward the instantaneous center of bending.

1.6 What it means to build a model

Geometry supplies identities. Mechanics supplies balances. Constitutive assumptions say what a material resists. Boundary conditions specify how the material is held and loaded. These ingredients must not be conflated. For example, "a freely sliding taut fiber on a smooth surface is a geodesic" follows from a mechanical assumption and a geometric definition. It does not assert that every biological fiber is a geodesic.

We will usually assume smooth fields, nonzero tension, and regular coordinate charts. At an edge, a straight segment, a pressure critical point, or a symmetry axis, some of these assumptions can fail. The correct response is to return to the vector equations and identify which divisions or normalizations remain legitimate.

Chapter 2

Curves, curvature, and tension

2.1 Arc length removes the arbitrary speed of a parametrization

A regular curve is a differentiable map $\mathbf{x}(t)$ with $\mathbf{x}'(t) \neq 0$. The same geometric path can be traced quickly, slowly, or with varying speed. Arc length measures the path itself:

$$s(t) = s(t_0) + \int_{t_0}^{t} |\mathbf{x}'(q)|\,\mathrm{d}q, \qquad \frac{\mathrm{d}s}{\mathrm{d}t} = |\mathbf{x}'(t)|. \tag{2.1}$$

Since this derivative is positive, s is locally an admissible parameter. By the chain rule,

$$\boldsymbol{\tau} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}s} = \frac{\mathbf{x}'(t)}{|\mathbf{x}'(t)|}, \qquad |\boldsymbol{\tau}| = 1. \tag{2.2}$$

The unit tangent describes direction without speed. Reversing the direction in which the curve is traversed reverses $\boldsymbol{\tau}$, but does not change the underlying unoriented curve.

2.2 Curvature is turning per unit distance

Differentiate $\boldsymbol{\tau} \cdot \boldsymbol{\tau} = 1$ with respect to s:

$$\boldsymbol{\tau} \cdot \frac{\mathrm{d}\boldsymbol{\tau}}{\mathrm{d}s} = 0.$$

The tangent therefore turns into its normal plane. Define

$$\boldsymbol{\kappa} = \frac{\mathrm{d}\boldsymbol{\tau}}{\mathrm{d}s}, \qquad \kappa = |\boldsymbol{\kappa}|. \tag{2.3}$$

The vector $\boldsymbol{\kappa}$ is the curvature vector; the scalar $\kappa \geq 0$ is the curvature. When $\kappa > 0$, the *principal normal* is

$$\mathbf{n}_c = \frac{\boldsymbol{\kappa}}{\kappa}, \qquad \frac{\mathrm{d}\boldsymbol{\tau}}{\mathrm{d}s} = \kappa \mathbf{n}_c. \tag{2.4}$$

The subscript distinguishes this curve normal from the surface normal $\mathbf{n}$ introduced later. There are infinitely many vectors perpendicular to $\boldsymbol{\tau}$; $\mathbf{n}_c$ is singled out by the actual bending of the curve.

The plane spanned by $\boldsymbol{\tau}$ and $\mathbf{n}_c$ is the osculating plane. A Taylor expansion gives its geometric meaning:

$$\mathbf{x}(s+h) = \mathbf{x}(s) + h\boldsymbol{\tau}(s) + \frac{h^2}{2}\kappa(s)\mathbf{n}_c(s) + O(h^3). \tag{2.5}$$

To second order, the curve lies in this plane. The circle matching its position, tangent, and curvature has radius $1/\kappa$; this is the osculating circle. If $\kappa = 0$, the curvature vector still exists, but this construction does not define a principal normal or a finite osculating circle.

Example 2.1 (A circle)**.** For $\mathbf{x}(s) = (R\cos(s/R), R\sin(s/R), 0)$,

$$\boldsymbol{\tau} = (-\sin(s/R), \cos(s/R), 0), \qquad \boldsymbol{\tau}_s = -\frac{1}{R}(\cos(s/R), \sin(s/R), 0).$$

Hence $\kappa = 1/R$, and the principal normal points inward. The curve's normal is not necessarily the outward normal of a surface containing it.

2.3 Curvature in a general parameter

Write $v = |\mathbf{x}_t|$, so that $\mathbf{x}_t = v\boldsymbol{\tau}$. Differentiation gives

$$\mathbf{x}_{tt} = v_t\boldsymbol{\tau} + v^2\kappa\mathbf{n}_c. \tag{2.6}$$

Acceleration has a tangential part that changes speed and a normal part that changes direction. Taking a cross product with $\mathbf{x}_t$ yields

$$|\mathbf{x}_t \times \mathbf{x}_{tt}| = v^3\kappa,$$

so

$$\boxed{\kappa = \frac{|\mathbf{x}_t \times \mathbf{x}_{tt}|}{|\mathbf{x}_t|^3}.} \tag{2.7}$$

This formula is useful computationally, but the definition $\kappa = |\boldsymbol{\tau}_s|$ is conceptually more informative.

2.4 The Frenet frame and torsion

For a sufficiently smooth curve with $\kappa > 0$, define the binormal $\mathbf{b} = \boldsymbol{\tau} \times \mathbf{n}_c$. The ordered triple $(\boldsymbol{\tau}, \mathbf{n}_c, \mathbf{b})$ is an orthonormal frame. Differentiating its pairwise inner products shows that its derivative matrix is skew-symmetric. Since $\boldsymbol{\tau}_s = \kappa\mathbf{n}_c$, the remaining derivatives have the form

$$\boldsymbol{\tau}_s = \kappa\mathbf{n}_c, \qquad (\mathbf{n}_c)_s = -\kappa\boldsymbol{\tau} + \tau_{\mathrm{F}}\mathbf{b}, \qquad \mathbf{b}_s = -\tau_{\mathrm{F}}\mathbf{n}_c. \tag{2.8}$$

Here τ_{F} is the *torsion of the space curve*; the subscript avoids confusing this scalar with the tangent vector $\boldsymbol{\tau}$. To see the coefficient $-\kappa$, differentiate $\mathbf{n}_c \cdot \boldsymbol{\tau} = 0$. To see that $\mathbf{b}_s$ has no tangent component, differentiate $\mathbf{b} \cdot \boldsymbol{\tau} = 0$. The last two coefficients must be negatives of each other because $\mathbf{b} \cdot \mathbf{n}_c = 0$.

Curvature measures bending of the tangent. Torsion measures rotation of the osculating plane. A planar curve has constant binormal and therefore zero torsion wherever its Frenet frame exists. For a general parameter,

$$\tau_{\mathrm{F}} = \frac{\det(\mathbf{x}_t, \mathbf{x}_{tt}, \mathbf{x}_{ttt})}{|\mathbf{x}_t \times \mathbf{x}_{tt}|^2}.$$

This follows by inserting $\mathbf{x}_t = v\boldsymbol{\tau}$ and (2.6) into the triple product: only the binormal part of $\mathbf{x}_{ttt}$ survives.

Example 2.2 (A circular helix)**.** Take $\mathbf{x}(t) = (R\cos t, R\sin t, bt)$. The speed is the constant $a = \sqrt{R^2 + b^2}$, so $s = at$. Direct differentiation gives

$$\kappa = \frac{R}{R^2 + b^2}, \qquad \tau_{\mathrm{F}} = \frac{b}{R^2 + b^2}, \qquad \mathbf{n}_c = (-\cos t, -\sin t, 0).$$

The helix bends radially inward even though it rises in the axial direction. This fact will explain why it is a geodesic on a cylinder.

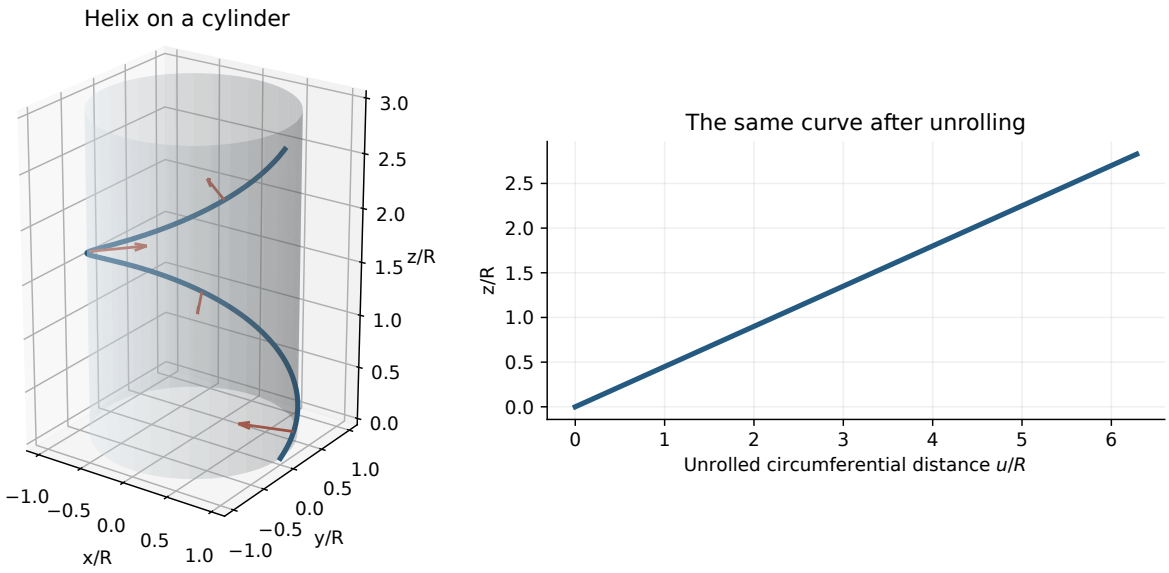


Figure 2.1: A helical fiber and its unrolled image. The red arrows point in the direction of the space-curve curvature. On the cylinder that direction is surface-normal, so the helix is a geodesic.

2.5 Deriving the cable equation from endpoint forces

Consider a short cable segment $[s, s + \Delta s]$. The rest of the cable pulls its right endpoint with force $T(s + \Delta s)\boldsymbol{\tau}(s + \Delta s)$ and its left endpoint with force $-T(s)\boldsymbol{\tau}(s)$. If $\mathbf{q}$ is the external force on the cable per unit arc length, equilibrium is

$$T(s + \Delta s)\boldsymbol{\tau}(s + \Delta s) - T(s)\boldsymbol{\tau}(s) + \int_s^{s+\Delta s} \mathbf{q}(\xi)\,\mathrm{d}\xi = 0.$$

Divide by Δs and take the limit:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}s}(T\boldsymbol{\tau}) + \mathbf{q} = 0.} \tag{2.9}$$

The force exerted by the cable *on its surroundings* is the opposite of $\mathbf{q}$:

$$\mathbf{f} = \frac{\mathrm{d}}{\mathrm{d}s}(T\boldsymbol{\tau}) = T_s\boldsymbol{\tau} + T\kappa\mathbf{n}_c. \tag{2.10}$$

Keeping track of which body receives a force prevents a common sign error. Equation (2.9) is force balance on the cable; equation (2.10) describes the cable's action on its constraints.

2.6 Free sliding and constant tension

Suppose a cable is constrained to a prescribed curve and can slide along it without friction. The constraint cannot exert a force in the tangent direction. Taking the dot product of (2.9) with $\boldsymbol{\tau}$ gives

$$T_s + \mathbf{q} \cdot \boldsymbol{\tau} = 0.$$

Thus $\mathbf{q} \cdot \boldsymbol{\tau} = 0$ implies

$$\boxed{T_s = 0, \qquad \mathbf{f} = T\kappa\mathbf{n}_c.} \tag{2.11}$$

Constant tension does not mean zero resultant force. The forces at two ends can have equal magnitudes but different directions. Their vector difference is precisely the curvature force.

For a circular cable, the force per length on its support is T/R, directed inward. Doubling the tension doubles the load; doubling the radius halves it. A straight cable of constant tension has zero distributed force in its interior, although it still pulls at its ends.

There are two distinct constraints we will use. If the cable is prescribed to lie on a *curve*, all transverse motions are forbidden. If it is prescribed only to lie on a *surface*, it can also move sideways within the tangent plane. In the second case a frictionless constraint supplies only surface-normal force. This stronger freedom will force the cable to follow a geodesic.

Exercise 2.1. Integrate $T\kappa\mathbf{n}_c$ along a circular arc subtending angle $2a$. Verify that the magnitude of the resultant is $2T \sin a$, agreeing with the difference of endpoint tension vectors.

Exercise 2.2. For a helical cable with radius R, rise parameter b, and constant tension T, find the force per unit length and the force per unit azimuthal angle. Explain why they have different units.

Chapter 3

Surfaces and their two fundamental forms

3.1 A surface as a regular parametrization

Let $U \subset \mathbb{R}^2$ be open. A smooth map $\mathbf{x}\colon U \to \mathbb{R}^3$ describes a regular surface patch if

$$\mathbf{x}_u \times \mathbf{x}_v \neq 0.$$

This condition says that the two coordinate directions remain independent. The tangent plane at $\mathbf{x}(u, v)$ is

$$T_{\mathbf{x}}S = \operatorname{span}\{\mathbf{x}_u, \mathbf{x}_v\}.$$

It consists of the velocity vectors of curves lying on the surface: equation (1.5) proves one inclusion, and a straight coordinate path with prescribed coordinate velocity proves the other.

An orientation is a continuous choice of unit normal. In a chart we choose

$$\mathbf{n} = \frac{\mathbf{x}_u \times \mathbf{x}_v}{|\mathbf{x}_u \times \mathbf{x}_v|}. \tag{3.1}$$

Reversing orientation changes $\mathbf{n}$ to $-\mathbf{n}$. All signed normal curvatures change sign with it, but Gaussian curvature does not.

3.2 The first fundamental form: measuring on the surface

For a small coordinate displacement,

$$\mathrm{d}\mathbf{x} = \mathbf{x}_u \mathrm{d}u + \mathbf{x}_v \mathrm{d}v.$$

Taking the squared Euclidean length gives

$$\boxed{\mathrm{d}s^2 = E\mathrm{d}u^2 + 2F\mathrm{d}u\mathrm{d}v + G\mathrm{d}v^2,} \tag{3.2}$$

where

$$E = \mathbf{x}_u \cdot \mathbf{x}_u, \qquad F = \mathbf{x}_u \cdot \mathbf{x}_v, \qquad G = \mathbf{x}_v \cdot \mathbf{x}_v. \tag{3.3}$$

The expression $\mathrm{d}s^2$ is a quadratic form, called the first fundamental form or metric. It assigns a squared length to each tangent displacement. It is not an equation saying that s is a globally defined coordinate.

Writing $g = (g_{ij}) = \left(\begin{smallmatrix} E & F \\ F & G \end{smallmatrix}\right)$, regularity is equivalent to

$$E > 0, \qquad EG - F^2 > 0.$$

Indeed, $EG - F^2 = |\mathbf{x}_u \times \mathbf{x}_v|^2$. Consequently

$$\mathrm{d}A = \sqrt{EG - F^2}\mathrm{d}u\mathrm{d}v. \tag{3.4}$$

A parameter rectangle need not be a physical rectangle: F measures the failure of coordinate directions to be orthogonal. Their angle ϕ satisfies

$$\cos\phi = \frac{F}{\sqrt{EG}}. \tag{3.5}$$

For a curve $(u(t), v(t))$, its length is

$$L = \int \sqrt{Eu_t^2 + 2Fu_tv_t + Gv_t^2}\,\mathrm{d}t.$$

For two tangent vectors with coordinate components a^i, b^i, their inner product is $g_{ij}a^ib^j$. Repeated indices in such expressions are summed over $1, 2$. Superscripts here label components; they do not mean powers.

Under new coordinates $q^i = q^i(\widetilde{q}^1, \widetilde{q}^2)$, the metric becomes

$$\widetilde{g}_{ab} = g_{ij}\frac{\partial q^i}{\partial \widetilde{q}^a}\frac{\partial q^j}{\partial \widetilde{q}^b}.$$

The coefficients change, but the length of a physical displacement does not. This is the simplest example of a tensor transformation law.

Example 3.1 (Cylinder versus plane)**.** Parametrize a cylinder of radius R by $\mathbf{x}(u, z) = (R\cos(u/R), R\sin(u/R), z)$. Then $E = G = 1$, $F = 0$, so $\mathrm{d}s^2 = \mathrm{d}u^2 + \mathrm{d}z^2$. A planar sheet rolled into a cylinder preserves all local lengths and angles. It changes its shape in space without changing its metric. This is an *isometry*, a distinction that becomes essential when discussing inextensible fiber networks.

3.3 The second fundamental form: bending out of the tangent plane

Second derivatives of $\mathbf{x}$ contain both tangential and normal information. Define

$$b_{11} = \mathbf{n} \cdot \mathbf{x}_{uu}, \qquad b_{12} = \mathbf{n} \cdot \mathbf{x}_{uv}, \qquad b_{22} = \mathbf{n} \cdot \mathbf{x}_{vv}. \tag{3.6}$$

The second fundamental form is

$$\mathrm{II} = b_{11}\mathrm{d}u^2 + 2b_{12}\mathrm{d}u\mathrm{d}v + b_{22}\mathrm{d}v^2. \tag{3.7}$$

Why does it measure curvature? For a unit-speed surface curve $\gamma(s) = \mathbf{x}(u(s), v(s))$,

$$\gamma_{ss} = \mathbf{x}_{uu}u_s^2 + 2\mathbf{x}_{uv}u_s v_s + \mathbf{x}_{vv}v_s^2 + \mathbf{x}_u u_{ss} + \mathbf{x}_v v_{ss}.$$

Taking its dot product with $\mathbf{n}$ eliminates the last two terms. Thus the signed normal curvature is

$$k_n = \gamma_{ss} \cdot \mathbf{n} = b_{11}u_s^2 + 2b_{12}u_s v_s + b_{22}v_s^2. \tag{3.8}$$

For a nonunit tangent direction with components (a, b), divide by its squared length:

$$k_n(a, b) = \frac{b_{11}a^2 + 2b_{12}ab + b_{22}b^2}{Ea^2 + 2Fab + Gb^2}. \tag{3.9}$$

All curves through a point with the same tangent direction have the same normal curvature there. They can bend differently within the surface, but the surface fixes the normal part of their bending.

3.4 The Gauss map and shape operator

The map

$$N\colon S \longrightarrow S^2, \qquad N(\mathbf{x}) = \mathbf{n}(\mathbf{x})$$

sends each point to its unit normal, now regarded as a point on the unit sphere. This is the *Gauss map*, called the spherical image in the original notes.

Because $\mathbf{n} \cdot \mathbf{n} = 1$, every derivative of $\mathbf{n}$ is perpendicular to $\mathbf{n}$, hence tangent to the surface. Define the shape operator

$$S(\mathbf{a}) = -\mathrm{D}_{\mathbf{a}}\mathbf{n}. \tag{3.10}$$

The minus sign fixes our convention. Differentiate $\mathbf{n} \cdot \mathbf{x}_j = 0$:

$$-\mathbf{n}_i \cdot \mathbf{x}_j = \mathbf{n} \cdot \mathbf{x}_{ij} = b_{ij}.$$

Therefore

$$\langle S\mathbf{x}_i, \mathbf{x}_j \rangle = b_{ij}, \qquad [S]^i{}_j = g^{ik}b_{kj}, \tag{3.11}$$

where $(g^{ij}) = g^{-1}$. Symmetry of b_{ij} shows that S is self-adjoint with respect to the metric. A real self-adjoint operator on a two-dimensional inner-product space has an orthonormal eigenbasis. Its eigenvalues k_1, k_2 are the *principal curvatures*; its eigenvectors are the principal directions.

For a unit tangent $\mathbf{a} = \cos\psi\,\mathbf{e}_1 + \sin\psi\,\mathbf{e}_2$ in principal directions,

$$\boxed{k_n(\psi) = k_1\cos^2\psi + k_2\sin^2\psi.} \tag{3.12}$$

This is Euler's normal-curvature formula: expand $\langle S\mathbf{a}, \mathbf{a}\rangle$ and use orthogonality. It is the precise reason that a saddle has directions of zero normal curvature.

Define Gaussian and mean curvature by

$$K = \det S = k_1 k_2 = \frac{b_{11}b_{22} - b_{12}^2}{EG - F^2}, \qquad H = \frac{1}{2}\operatorname{tr} S = \frac{k_1 + k_2}{2}. \tag{3.13}$$

The letter K will always mean Gaussian curvature in these notes. We will use C for the integration constant called K in the final pages of the source.

3.5 Examples and the sign of curvature

Example 3.2 (Sphere)**.** On a sphere of radius R with outward normal, $\mathbf{n} = \mathbf{x}/R$, so $S = -\mathrm{Id}/R$. Thus $k_1 = k_2 = -1/R$, $H = -1/R$, and $K = 1/R^2$. Negative principal curvatures here reflect the outward normal convention; the sphere still has positive Gaussian curvature.

Example 3.3 (Cylinder)**.** For a cylinder with outward radial normal, the circumferential principal curvature is $-1/R$, and the axial principal curvature is zero. Hence $K = 0$, even though the cylinder bends in space. The normal remains constant as one moves along a generator, so its Gauss map collapses that direction.

Example 3.4 (A local saddle)**.** For the graph $\mathbf{x}(u, v) = (u, v, (au^2 + bv^2)/2)$, at the origin with upward normal, $g = \mathrm{Id}$ and $b = \mathrm{diag}(a, b)$. Thus $K = ab$. If $a > 0 > b$, there are two unoriented directions with $\tan^2 \psi = -a/b$ and $k_n = 0$.

The Gauss map has differential $\mathrm{d}N = -S$. In two dimensions $\det(-S) = \det S = K$. Consequently the pullback of oriented spherical area is

$$\boxed{N^*(\,\mathrm{d}A_{S^2}) = K\,\mathrm{d}A.} \tag{3.14}$$

The star means that we measure the image area using the original surface coordinates. Positive K preserves orientation, negative K reverses it, and zero K collapses area to first order. Ordinary unsigned image area has density $|K|$, and overlaps must be counted with multiplicity. These qualifications matter: "Gaussian curvature equals spherical-image area divided by area" is a *signed local Jacobian* statement, not a claim about the unsigned area of an arbitrary image set.

3.6 Splitting the curvature of a surface curve

Along a unit-speed surface curve set

$$\mathbf{m} = \mathbf{n} \times \boldsymbol{\tau}.$$

Then $(\boldsymbol{\tau}, \mathbf{m}, \mathbf{n})$ is an orthonormal frame. Since $\boldsymbol{\tau}_s \perp \boldsymbol{\tau}$,

$$\boxed{\boldsymbol{\tau}_s = k_g \mathbf{m} + k_n \mathbf{n}, \qquad \kappa^2 = k_g^2 + k_n^2.} \tag{3.15}$$

The scalar $k_g = \boldsymbol{\tau}_s \cdot \mathbf{m}$ is the signed *geodesic curvature*. It measures turning visible to an observer confined to the surface. The normal curvature k_n measures turning out of the tangent plane.

A geodesic has $k_g = 0$. An asymptotic curve has $k_n = 0$. These are different conditions. On a sphere, a great circle is geodesic but has nonzero normal curvature. On a plane, every curve has zero normal curvature, while only straight lines are geodesics. A curve satisfying both conditions is locally a straight line in ambient space.

Exercise 3.1. Use (3.12) to show that an asymptotic direction can exist only if $K \leq 0$. Describe separately the cases $K < 0$, $K = 0$ with one nonzero principal curvature, and $S = 0$.

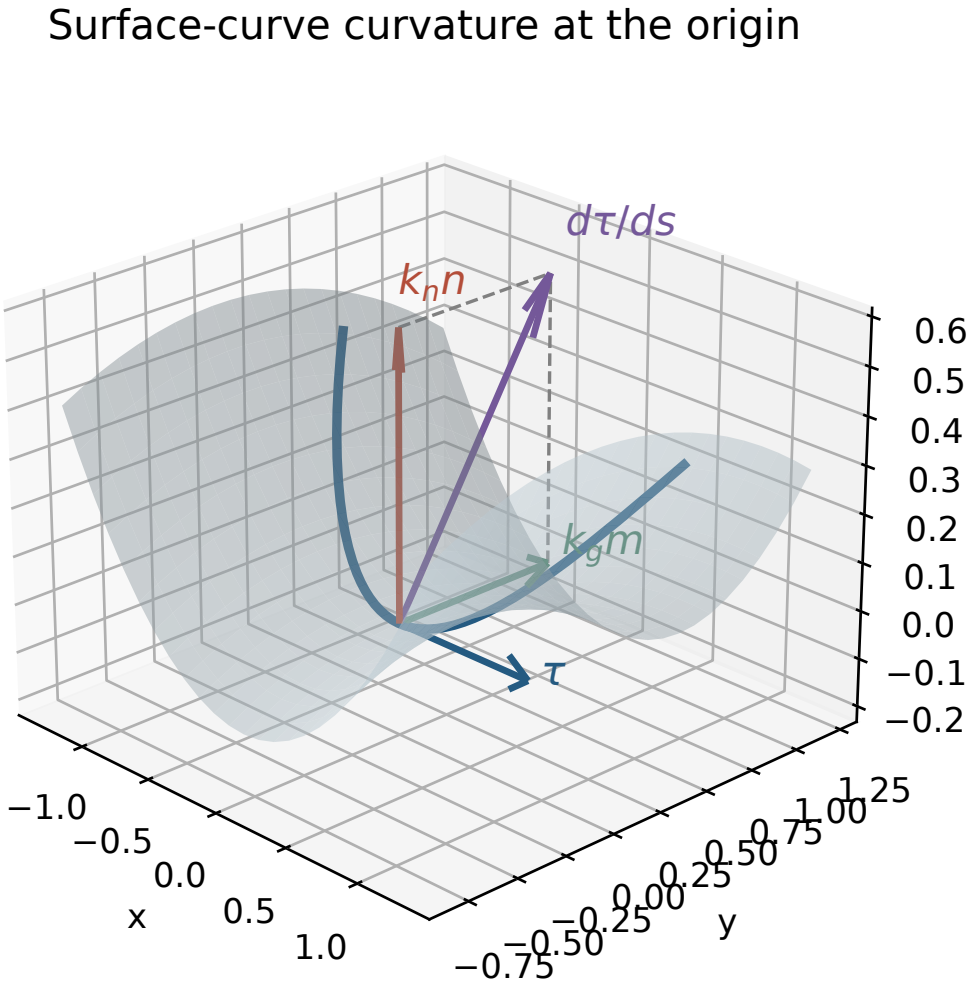


Figure 3.1: At the origin of this surface patch, the curvature vector splits into a tangent contribution $k_g\mathbf{m}$ and a normal contribution $k_n\mathbf{n}$. The blue curve lies on the surface. The arrows show vectors at one point, not a second curve.

Chapter 4

Parallel transport, holonomy, and Gaussian curvature

4.1 Why ordinary constancy does not work on a curved surface

Two vectors at different points in $\mathbb{R}^3$ can be compared by ordinary translation. But a vector tangent at one point of a curved surface will generally cease to be tangent after that translation. We need a notion of "no turning" that respects the changing tangent planes.

Let $\mathbf{w}(s)$ be a tangent vector field along a surface curve $\gamma(s)$. Its ambient derivative $\mathbf{w}_s$ need not be tangent. Define the covariant derivative along the curve to be its tangential projection:

$$\frac{\mathrm{D}\mathbf{w}}{\mathrm{d}s} = P\mathbf{w}_s, \qquad P = \mathrm{Id} - \mathbf{n} \otimes \mathbf{n}. \tag{4.1}$$

Here $(\mathbf{n} \otimes \mathbf{n})\mathbf{a} = \mathbf{n}(\mathbf{n} \cdot \mathbf{a})$, so P removes the normal component. This construction gives the Levi-Civita connection of the induced metric. The word *connection* expresses that it connects, or compares, tangent spaces at nearby points.

Definition 4.1. A tangent vector field is parallel along a curve if $\mathrm{D}\mathbf{w}/\mathrm{d}s = 0$.

Thus a parallel field may change as an ambient vector, but only in the normal direction. This change is exactly what is needed to remain tangent.

4.2 Deriving Peskin's transport equation

Parallelism says $\mathbf{w}_s = a(s)\mathbf{n}$, while tangency says $\mathbf{w} \cdot \mathbf{n} = 0$. Differentiate the latter:

$$0 = \mathbf{w}_s \cdot \mathbf{n} + \mathbf{w} \cdot \mathbf{n}_s = a + \mathbf{w} \cdot \mathbf{n}_s.$$

Therefore

$$\boxed{\mathbf{w}_s = -(\mathbf{w} \cdot \mathbf{n}_s)\mathbf{n}.} \tag{4.2}$$

This is a linear ordinary differential equation for $\mathbf{w}$, once the curve and its normal field are known. If the initial vector is tangent, the equation preserves tangency because the derivative of $\mathbf{w} \cdot \mathbf{n}$ is identically zero. With smooth coefficients, the initial-value problem has a unique solution. It is equally valid in any regular curve parameter if every derivative is taken in that parameter.

If $\mathbf{w}_1$ and $\mathbf{w}_2$ are parallel along the same curve, then

$$\frac{\mathrm{d}}{\mathrm{d}s}(\mathbf{w}_1 \cdot \mathbf{w}_2) = -(\mathbf{w}_1 \cdot \mathbf{n}_s)(\mathbf{n} \cdot \mathbf{w}_2) - (\mathbf{w}_2 \cdot \mathbf{n}_s)(\mathbf{w}_1 \cdot \mathbf{n}) = 0.$$

Setting $\mathbf{w}_1 = \mathbf{w}_2$ proves preservation of length; the general case proves preservation of angles. Parallel transport is thus a linear isometry between the initial and final tangent planes. Reversing the path reverses the transport map by uniqueness of the ODE.

4.3 Why the spherical image gives the same transport

The tangent plane of the unit sphere at the point $\mathbf{n}(s)$ consists of vectors perpendicular to $\mathbf{n}(s)$. As a vector subspace of $\mathbb{R}^3$, this is exactly the tangent plane of the original surface at $\gamma(s)$. The sphere's outward normal at $\mathbf{n}(s)$ is again $\mathbf{n}(s)$. Therefore the transport equation on the original curve and on its spherical image is the same equation (4.2).

This comparison identifies tangent vectors by ordinary translation in $\mathbb{R}^3$. It does *not* send a vector through the differential $\mathrm{d}N$, which generally changes its length. Also, s need not be arc length on the image curve. Parameter invariance makes that irrelevant. Even if the image momentarily stops, the ODE still makes sense.

4.4 Holonomy: the rotation left after a round trip

For a closed curve, parallel transport returns to the original tangent plane. It preserves lengths, angles, and orientation, so its net action is a planar rotation. Its signed angle $\beta \in \mathbb{R}/(2\pi\mathbb{Z})$ is the *holonomy angle*. It is independent of the initial nonzero vector. Changing the starting point conjugates the map by a transport isometry; because rotations in an oriented two-dimensional plane commute, the angle is unchanged.

On a plane, $\mathbf{n}_s = 0$, so a parallel vector is constant in $\mathbb{R}^3$, and every closed loop has zero holonomy. On a sphere, transport around a loop generally produces a nonzero rotation.

Example 4.2 (A spherical triangle with two meridians)**.** On the unit sphere, travel from the north pole down one meridian to the equator, east along the equator through angle a, and back to the pole along the second meridian. Start with a vector pointing south. Along the meridians, the south-pointing unit field changes only normally; along the equator it is the fixed vector $(0, 0, -1)$. The final vector points south along the second meridian, making angle a with the initial vector. The enclosed spherical wedge has area

$$\int_0^a \int_0^{\pi/2} \sin\vartheta \, \mathrm{d}\vartheta \, \mathrm{d}\varphi = a.$$

Thus $\beta = a$, with the indicated positive orientation. On a sphere of radius R, the area is $R^2 a$, and $K = 1/R^2$, so the same result is $\beta = \int K \, \mathrm{d}A$.

4.5 An explicit derivation of the curvature–holonomy relation

We now justify the general formula rather than infer it from additivity alone. Work on an oriented patch that is topologically a disk. Choose a smooth orthonormal tangent frame $(\mathbf{e}_1, \mathbf{e}_2)$ with

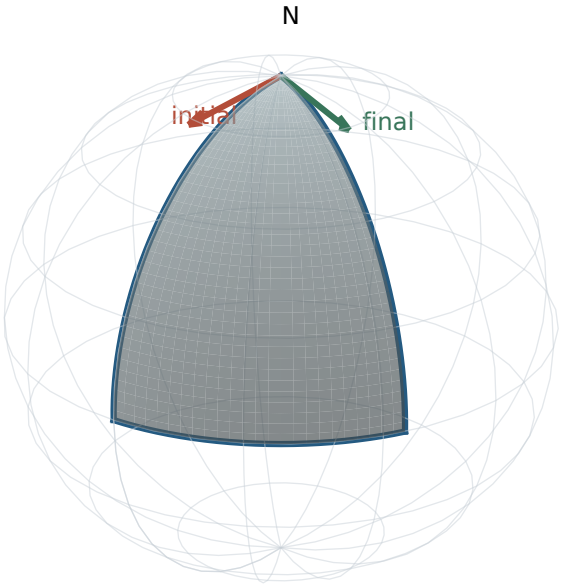


Figure 4.1: A spherical loop formed by two meridians and an equatorial arc. The initial and final transported vectors at the north pole differ by the angle between the meridians. For the unit sphere, that angle equals the shaded area.

$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{n}$. For a tangent direction X, define

$$\omega(X) = \langle \mathrm{D}_X \mathbf{e}_1, \mathbf{e}_2 \rangle.$$

This scalar tells us how quickly the chosen frame rotates in the tangent plane. As a function linear in X, ω is a differential one-form. In coordinates it has the elementary form $A\mathrm{d}u + B\mathrm{d}v$.

Write a unit tangent vector as $\mathbf{w} = \cos\theta\,\mathbf{e}_1 + \sin\theta\,\mathbf{e}_2$. Since $\mathrm{D}_X\mathbf{e}_1 = \omega(X)\mathbf{e}_2$ and $\mathrm{D}_X\mathbf{e}_2 = -\omega(X)\mathbf{e}_1$, differentiation yields

$$\frac{\mathrm{D}\mathbf{w}}{\mathrm{d}s} = (\theta_s + \omega(\gamma_s))(-\sin\theta\,\mathbf{e}_1 + \cos\theta\,\mathbf{e}_2). \tag{4.3}$$

Hence parallel transport is exactly

$$\theta_s = -\omega(\gamma_s), \qquad \beta = -\oint_{\partial D} \omega \pmod{2\pi}. \tag{4.4}$$

To connect frame rotation with surface bending, let $\eta_i(X) = \langle \mathrm{D}^{\mathbb{R}^3}_X \mathbf{e}_i, \mathbf{n} \rangle$. The ambient derivative identities are

$$\begin{aligned} \mathrm{d}\mathbf{e}_1 &= \omega\mathbf{e}_2 + \eta_1\mathbf{n}, \\ \mathrm{d}\mathbf{e}_2 &= -\omega\mathbf{e}_1 + \eta_2\mathbf{n}, \\ \mathrm{d}\mathbf{n} &= -\eta_1\mathbf{e}_1 - \eta_2\mathbf{e}_2. \end{aligned}$$

For completeness, the exterior product of two one-forms is the alternating expression $(\alpha\wedge\eta)(X,Y) = \alpha(X)\eta(Y) - \alpha(Y)\eta(X)$. For $\omega = A\mathrm{d}u + B\mathrm{d}v$, its exterior derivative is $\mathrm{d}\omega = (B_u - A_v)\mathrm{d}u \wedge \mathrm{d}v$. These definitions package ordinary mixed-derivative calculations.

Since ambient mixed derivatives commute, differentiating the first frame identity again and taking its $\mathbf{e}_2$ component gives

$$0 = \mathrm{d}\omega + \eta_1 \wedge \eta_2.$$

Evaluate the last wedge on an oriented orthonormal pair. Its value is the determinant of the second fundamental form in that frame, namely K. Therefore

$$\mathrm{d}\omega = -K\,\mathrm{d}A. \tag{4.5}$$

Green's theorem, equivalently Stokes' theorem for one-forms, gives

$$\boxed{\beta = \int_D K\,\mathrm{d}A \pmod{2\pi}.} \tag{4.6}$$

This also explains subdivision: integrals over interior boundaries cancel with opposite orientations. A real-valued angle can be chosen continuously for small loops and added consistently; the observable rotation of a vector always remains defined modulo 2π.

4.6 Intrinsic curvature and Gauss–Bonnet

The tangential projection connection can be computed entirely from the metric (Chapter 5). Equation (4.5) therefore shows that K, although defined using the embedding and its normal, is determined by intrinsic length measurements. This is the content of Gauss's *Theorema Egregium*. Rolling a sheet into a cylinder preserves $K = 0$; bending it into a spherical patch with $K > 0$ requires a change of metric somewhere.

For a positively oriented smooth boundary, its unit tangent has an angle θ in the chosen frame, and $k_g = \theta_s + \omega(\gamma_s)$. Around a simple disk boundary the tangent turns once relative to a smooth disk frame. At corners add the signed exterior turning angles ϵ_j. Thus

$$\int_{\partial D} k_g\,\mathrm{d}s + \sum_j \epsilon_j = 2\pi + \oint_{\partial D} \omega.$$

Combining this with (4.5) gives the disk form of Gauss–Bonnet:

$$\boxed{\int_D K\,\mathrm{d}A + \int_{\partial D} k_g\,\mathrm{d}s + \sum_j \epsilon_j = 2\pi.} \tag{4.7}$$

For a geodesic triangle with interior angles A, B, C, the boundary integral vanishes and the exterior angles are $\pi - A, \pi - B, \pi - C$. Hence

$$A + B + C - \pi = \int_D K\,\mathrm{d}A.$$

On a sphere, triangle angle excess measures area. On a flat plane, it vanishes.

For a more general compact region, the right side is $2\pi\chi(D)$, where χ is the Euler characteristic, computed from a triangulation as vertices minus edges plus faces. This follows by applying the disk formula to pieces and canceling shared edges. A loop that does not bound a disk requires appropriate topological care; the local curvature formula should not be used to invent an enclosed disk that does not exist.

Chapter 5

Geodesics: straightness within a surface

5.1 The geometric and mechanical definitions coincide

A unit-speed surface curve is a geodesic when its tangent is parallel along itself:

$$\frac{\mathrm{D}\boldsymbol{\tau}}{\mathrm{d}s} = 0. \tag{5.1}$$

By (3.15), this is exactly $k_g = 0$, or

$$\mathbf{x}_{ss} = k_n \mathbf{n}.$$

The signed coefficient k_n can be positive, negative, or zero. Where $\kappa > 0$, the principal normal is $\mathbf{n}_c = \pm\mathbf{n}$, depending on orientation. It is safer to say that the curvature vector is surface-normal than to insist that the two unit normals are equal.

Now place a taut cable on a smooth surface, allowing it to slide freely in every tangent direction. The external constraint force is normal to the surface. Tangentially projecting the cable balance gives

$$T_s \boldsymbol{\tau} + T\frac{\mathrm{D}\boldsymbol{\tau}}{\mathrm{d}s} = 0.$$

Dotting with $\boldsymbol{\tau}$ proves $T_s = 0$. The remaining equation gives $\mathrm{D}\boldsymbol{\tau}/\mathrm{d}s = 0$ when $T > 0$. Thus a frictionlessly constrained cable is a geodesic. Its shape may be curved in ambient space because the supporting surface supplies a normal reaction.

5.2 Christoffel symbols are derivatives of a coordinate basis

Let $q^1 = u, q^2 = v$, and write $\mathbf{x}_i = \partial_i \mathbf{x}$. Since $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{n})$ is a basis, decompose the second derivatives as

$$\mathbf{x}_{ij} = \Gamma^k_{ij}\mathbf{x}_k + b_{ij}\mathbf{n}. \tag{5.2}$$

The coefficients Γ^k_{ij} are the Christoffel symbols. They describe how the tangent coordinate basis changes within the tangent plane. They are not extra physical forces or new material parameters.

To compute them from the metric, differentiate $g_{ij} = \mathbf{x}_i \cdot \mathbf{x}_j$:

$$\partial_i g_{j\ell} = \mathbf{x}_{ij} \cdot \mathbf{x}_\ell + \mathbf{x}_j \cdot \mathbf{x}_{i\ell}.$$

Add this equation to the one with i, j interchanged, then subtract $\partial_\ell g_{ij}$. Symmetry of mixed partials leaves

$$\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij} = 2\mathbf{x}_{ij} \cdot \mathbf{x}_\ell = 2\Gamma^k_{ij} g_{k\ell}.$$

Multiplying by the inverse metric gives

$$\boxed{\Gamma^k_{ij} = \frac{1}{2} g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).} \tag{5.3}$$

This calculation proves that the connection depends only on the first fundamental form.

For a tangent field $\mathbf{w} = w^i \mathbf{x}_i$ along $q(s)$, differentiate and use (5.2):

$$\frac{\mathrm{D}\mathbf{w}}{\mathrm{d}s} = \left(\frac{\mathrm{d}w^k}{\mathrm{d}s} + \Gamma^k_{ij} \frac{\mathrm{d}q^i}{\mathrm{d}s} w^j \right) \mathbf{x}_k. \tag{5.4}$$

For the tangent itself, $w^k = q^k_s$. The geodesic equation becomes

$$\boxed{q^k_{ss} + \Gamma^k_{ij} q^i_s q^j_s = 0, \quad k = 1, 2.} \tag{5.5}$$

This is a second-order ODE. A starting point and starting velocity determine a local geodesic uniquely in a smooth regular chart. Its speed stays constant because $\frac{\mathrm{d}}{\mathrm{d}s}\langle \boldsymbol{\tau}, \boldsymbol{\tau} \rangle = 2\langle \mathrm{D}\boldsymbol{\tau}/\mathrm{d}s, \boldsymbol{\tau} \rangle = 0$. Arc length corresponds to choosing unit initial speed.

5.3 Why geodesics are stationary-length curves

Consider a family of curves $\gamma_\epsilon(t)$ on the surface with fixed endpoints. Its variation field $V = \partial_\epsilon \gamma_\epsilon|_0$ is tangent and vanishes at the endpoints. For the energy

$$\mathcal{E}[\gamma] = \frac{1}{2} \int_a^b |\gamma_t|^2 \, \mathrm{d}t,$$

differentiation under the integral and integration by parts give

$$\begin{aligned} \delta\mathcal{E} &= \int_a^b \gamma_t \cdot V_t \, \mathrm{d}t \\ &= [\gamma_t \cdot V]_a^b - \int_a^b \gamma_{tt} \cdot V \, \mathrm{d}t \\ &= -\int_a^b \left\langle \frac{\mathrm{D}\gamma_t}{\mathrm{d}t}, V \right\rangle \mathrm{d}t. \end{aligned}$$

Since all tangent variations are allowed, stationarity implies $\mathrm{D}\gamma_t/\mathrm{d}t = 0$. The same calculation for length, taking a unit-speed reference curve, gives $\delta L = -\int \langle \mathrm{D}\boldsymbol{\tau}/\mathrm{d}s, V \rangle \, \mathrm{d}s$. A sufficiently short geodesic segment minimizes length between its endpoints. A long segment need not do so. On a sphere, a great-circle arc longer than a semicircle is still geodesic but is not a shortest route between its endpoints.

Energy and length are related by Cauchy–Schwarz:

$$L^2 = \left(\int_a^b |\gamma_t| \, \mathrm{d}t \right)^2 \le (b-a) \int_a^b |\gamma_t|^2 \, \mathrm{d}t = 2(b-a)\mathcal{E},$$

with equality exactly for constant speed. This explains why energy is often easier to vary while giving the same unparametrized geodesics.

5.4 Three complete examples

Example 5.1 (Plane)**.** Cartesian coordinates have constant metric and zero Christoffel symbols. Equation (5.5) reduces to $u_{ss} = v_{ss} = 0$, so the geodesics are straight lines.

Example 5.2 (Cylinder)**.** In the unrolled coordinates of Chapter 3, the metric is also $\mathrm{d}u^2 + \mathrm{d}z^2$. Hence $u = as + u_0$, $z = bs + z_0$, with $a^2 + b^2 = 1$. Rolling this line onto the cylinder gives

$$\mathbf{x}(s) = \left(R\cos\frac{as+u_0}{R}, R\sin\frac{as+u_0}{R}, bs + z_0 \right).$$

These are helices, including circles when $b = 0$ and generators when $a = 0$. Their curvature vector is radial and therefore surface-normal. A helix is curved in space and intrinsically straight on the cylinder.

Example 5.3 (Sphere)**.** On a radius-R sphere, a unit-speed geodesic has $\mathbf{x}_{ss}$ parallel to $\mathbf{x}$. Differentiating $\mathbf{x}\cdot\mathbf{x} = R^2$ twice gives $1 + \mathbf{x}\cdot\mathbf{x}_{ss} = 0$, so

$$\mathbf{x}_{ss} = -\frac{1}{R^2}\mathbf{x}.$$

Its solution is $\mathbf{x}(s) = \mathbf{a}\cos(s/R) + \mathbf{b}\sin(s/R)$, with $|\mathbf{a}| = |\mathbf{b}| = R$ and $\mathbf{a}\cdot\mathbf{b} = 0$. It lies in a plane through the sphere's center, and therefore traces a great circle. A smaller latitude circle is not geodesic.

5.5 Geodesics on surfaces of revolution and Clairaut's relation

Write a surface of revolution in meridional arc length ℓ:

$$\mathbf{x}(\ell,\theta) = (r(\ell)\cos\theta, r(\ell)\sin\theta, z(\ell)), \qquad r_\ell^2 + z_\ell^2 = 1.$$

Its metric is

$$\mathrm{d}s^2 = \mathrm{d}\ell^2 + r(\ell)^2\mathrm{d}\theta^2.$$

The energy Lagrangian is $\frac{1}{2}(\ell_s^2 + r^2\theta_s^2)$. It does not depend explicitly on θ. The corresponding Euler–Lagrange equation is therefore

$$\frac{\mathrm{d}}{\mathrm{d}s}(r^2\theta_s) = 0, \qquad r^2\theta_s = L. \tag{5.6}$$

If α is the angle between the fiber tangent and the positive circumferential direction, then $r\theta_s = \cos\alpha$, so

$$r\cos\alpha = L. \tag{5.7}$$

Some books measure the angle from the meridian and write a sine instead. The content is identical once the angle convention is fixed.

Since $|\cos\alpha| \le 1$, a geodesic with $L \neq 0$ cannot reach $r < |L|$, and in particular cannot reach the symmetry axis. This elementary restriction will have a major consequence for the proposed pressure surfaces of the heart wall.

Exercise 5.1. On a sphere of radius R, calculate k_n and $|k_g|$ for a latitude at colatitude ϑ. Check that the equator is the only nondegenerate latitude that is a geodesic.

Chapter 6

The geometry of heart-valve leaflets

6.1 A thin leaflet as a tension-bearing membrane

A membrane is an idealization of a thin sheet that supports in-plane tension but has negligible bending resistance. A valve leaflet under a pressure difference is a natural setting for this model. The pressure acts normally; the internal tensions act tangentially across cuts through the sheet. Equilibrium converts these different directions into restrictions on curvature.

It is useful to think first about a small membrane patch. Equal tangent forces on opposite sides cancel only if their directions are equal. If the surface curves, those directions differ, leaving a normal resultant. This is the surface analogue of the cable law $T\kappa\mathbf{n}_c$.

If the membrane resultant tensor is $\mathcal{N}$, with units force/length, its normal force balance is

$$\mathcal{N} : \mathrm{II} + p = 0, \tag{6.1}$$

where the colon means contraction in an orthonormal tangent basis: $\mathcal{N} : \mathrm{II} = \sum_{a,b} \mathcal{N}_{ab} b_{ab}$. To see the origin of this expression, let the tensor carry traction $\mathcal{N}\nu$ across a tangent cut with unit conormal ν. Differentiating the tangent basis in the surface divergence contributes a normal term $\sum_{a,b} \mathcal{N}_{ab} b_{ab}$; derivatives of the scalar coefficients contribute tangential terms. Adding pressure gives (6.1).

When tension is isotropic, $\mathcal{N} = N\mathrm{Id}_{TS}$, the normal balance is $2NH + p = 0$. On a sphere with outward normal, $H = -1/R$, and outward pressure gives $p = 2N/R$. On a cylinder with circumferential tension N, the corresponding relation is $p = N/R$. These familiar Laplace laws are special cases of curvature converting tangential tension into normal load. An anisotropic leaflet need not satisfy an isotropic Laplace law.

6.2 The line of closure is an asymptotic curve

Suppose a leaflet is supported along an edge cable $\gamma(s)$, parametrized by arc length. Let $T_e(s) > 0$ be its tension, and let $\mathbf{f}_m(s)$ be the force per length exerted by the membrane on the cable. For an ideal membrane, this force lies in the tangent plane of the leaflet. The edge equation is

$$\frac{\mathrm{d}}{\mathrm{d}s}(T_e\boldsymbol{\tau}) + \mathbf{f}_m = 0. \tag{6.2}$$

Take the dot product with the surface normal:

$$T_{e,s}\underbrace{\boldsymbol{\tau}\cdot\mathbf{n}}_{0}+T_e\boldsymbol{\tau}_s\cdot\mathbf{n}+\underbrace{\mathbf{f}_m\cdot\mathbf{n}}_{0}=0.$$

Since $T_e>0$,

$$\boxed{k_n=0 \quad \text{along the supported edge.}} \tag{6.3}$$

Thus the line is asymptotic on the surface. Its curvature vector lies in the surface tangent plane. Notice how this differs from the freely sliding fiber problem: there the external force was normal to the surface and gave $k_g=0$; here the applied membrane force is tangent and gives $k_n=0$.

Euler's formula now proves that $K\le 0$ at such an edge. If both principal curvatures had the same nonzero sign, every normal curvature would have that sign and none could vanish. At a strictly saddle-shaped point, there are two asymptotic directions. At a parabolic point there is a zero-principal-curvature direction. At a flat point all tangent directions are asymptotic.

6.3 Why two supported leaflets meet tangentially

Suppose two leaflets share a closure curve and each separately satisfies the edge model. At a point where the curve has $\kappa>0$, each tangent plane contains both $\boldsymbol{\tau}$ and $\boldsymbol{\tau}_s$, because its normal curvature vanishes. These two vectors are linearly independent and span the curve's osculating plane. Therefore both surface tangent planes equal that osculating plane, and the leaflets meet tangentially.

The nonzero-curvature condition is essential to this proof. If the common edge is straight, $\boldsymbol{\tau}_s=0$, so the edge supplies only one tangent direction; two different planes can contain that same line. Also, added normal contact forces, bending moments, or other supports change the edge equation and can change the conclusion. Tangential meeting is a consequence of the stated idealized force balance.

6.4 Why the free edge is mechanically demanding

Let p be a typical pressure difference, R a typical leaflet curvature radius, r a typical edge curvature radius, and h the leaflet thickness. Interior normal balance gives the scale

$$\frac{N}{R}\sim p, \qquad N\sim pR.$$

Thus the interior material stress is

$$\sigma_{\text{interior}}\sim\frac{pR}{h}. \tag{6.4}$$

If the membrane pulls on the edge with load per length of order N, a curved cable balances it with $T_e/r\sim N$. Therefore

$$T_e\sim pRr.$$

If the edge's effective cross-sectional area is of order h^2, then

$$\sigma_{\text{edge}}\sim\frac{pRr}{h^2}, \qquad \frac{\sigma_{\text{edge}}}{\sigma_{\text{interior}}}\sim\frac{r}{h}. \tag{6.5}$$

For a thin leaflet, r/h can be large. The estimate explains two geometric ways to support the edge: increase the cable cross-sectional area, or decrease the edge radius by dividing it into shorter, sharply curved arches with appropriate supports. In Peskin's discussion these are associated with the thickened free edge of the aortic valve and the arch-like structure supported by the mitral chordae. Equation (6.5) is a scaling argument; its numerical factors and the actual edge area depend on anatomy and loading.

6.5 A leaflet carried by one family of fibers

Introduce coordinates (u, v) so that v labels a fiber and u is arc length along it. Then

$$\mathbf{x}_u = \boldsymbol{\tau}, \qquad |\boldsymbol{\tau}| = 1. \tag{6.6}$$

Let $\mathcal{T}(u, v)\mathrm{d}v$ be the total tension carried by the fiber strip between labels v and $v + \mathrm{d}v$. This definition matters: $\mathcal{T}$ is tension *per unit fiber label*, not necessarily tension per unit physical width.

The oriented area vector of a coordinate patch is $\boldsymbol{\tau} \times \mathbf{x}_v \mathrm{d}u\mathrm{d}v$, so its pressure load is

$$p(\boldsymbol{\tau} \times \mathbf{x}_v)\mathrm{d}u\mathrm{d}v.$$

The difference of tensions on the two fiber-normal sides is

$$\partial_u(\mathcal{T}\boldsymbol{\tau})\mathrm{d}u\mathrm{d}v.$$

Assuming no other load or transverse tension, equilibrium gives

$$\partial_u(\mathcal{T}\boldsymbol{\tau}) + p(\boldsymbol{\tau} \times \mathbf{x}_v) = 0. \tag{6.7}$$

Dot with $\boldsymbol{\tau}$. Since $\boldsymbol{\tau}_u \perp \boldsymbol{\tau}$ and a cross product is perpendicular to each factor,

$$\mathcal{T}_u = 0, \qquad \mathcal{T} = \mathcal{T}(v).$$

Therefore the governing system is

$$\boxed{\mathbf{x}_u = \boldsymbol{\tau}, \qquad \boldsymbol{\tau}_u = \frac{p}{\mathcal{T}(v)}(\mathbf{x}_v \times \boldsymbol{\tau}).} \tag{6.8}$$

The right side is surface-normal, so every fiber is a geodesic. Moreover, $\partial_u|\boldsymbol{\tau}|^2 = 0$, so a unit initial tangent remains unit.

Let $a = |\boldsymbol{\tau} \times \mathbf{x}_v| = \sqrt{G - F^2}$, using $E = 1$. The physical spacing perpendicular to the fiber is $a\mathrm{d}v$. Thus the membrane resultant in the fiber direction is $N_f = \mathcal{T}/a$. Dotting (6.7) with the chosen unit normal gives

$$\mathcal{T}k_n + pa = 0, \qquad N_f k_n + p = 0. \tag{6.9}$$

This identifies the physical tension hidden in the label-based formula.

6.6 What the generating equations determine

One may prescribe an initial transverse curve $\mathbf{x}(0, v) = \mathbf{c}(v)$, initial unit fiber directions $\boldsymbol{\tau}(0, v) = \mathbf{a}(v)$, and a positive strip tension $\mathcal{T}(v)$. Regularity initially requires $\mathbf{a} \times \mathbf{c}' \neq 0$. The pressure load and (6.8) then describe how the surface is to continue along the fibers.

These are partial differential equations: the evolution in u involves $\mathbf{x}_v$. A formal generating prescription does not by itself prove global existence, uniqueness for arbitrary data, or absence of fiber crossings and coordinate degeneracies. The equations remain a local model subject to compatibility and boundary conditions. Reversing the pressure direction reverses the normal bending, explaining how two leaflets can be generated away from a shared initial curve with opposite load signs.

Example 6.1 (A cylindrical one-family membrane)**.** Take $\mathbf{x}(u, v) = (R\cos(u/R), R\sin(u/R), v)$, with outward normal. Then $a = 1$, $k_n = -1/R$, and the fibers are circumferential geodesics. Equation (6.9) gives $\mathcal{T} = pR$, recovering the cylindrical membrane law directly from the fiber-patch balance.

Exercise 6.1. A membrane has $p = 1$ force/area unit, $R = 10$ length units, $r = 2$, and $h = 0.1$. Compute the two stress scales. Then replace the edge area h^2 by $10h^2$ and calculate the new ratio.

Chapter 7

Inextensible fibers and the geometry of leaflet motion

7.1 Material coordinates are labels carried by the tissue

Let $\mathbf{x}(u, v, t)$ describe a moving surface. If (u, v) identifies the same material point at every time, these are material coordinates. The time-dependent metric is

$$g(t) = \begin{pmatrix} E(u,v,t) & F(u,v,t) \\ F(u,v,t) & G(u,v,t) \end{pmatrix}.$$

For a material curve $(u(a), v(a))$, its length at time t is

$$L(t) = \int \sqrt{Eu_a^2 + 2Fu_a v_a + Gv_a^2}\, \mathrm{d}a.$$

A fiber is inextensible if every sufficiently small material segment of it keeps its length. Requiring only the total length of a whole fiber to stay fixed would be weaker: some segments could stretch while others shrink. We use the local, segmentwise condition.

The no-slip condition at intersections of fiber families ensures that the network really supplies fixed material labels. Without it, a coordinate grid attached to one family need not track the same points of a second family, and the metric argument below would no longer follow.

7.2 One family fixes one metric coefficient

Suppose the fibers are $v =$ constant, and choose u to be material arc length along them initially. Their inextensibility gives

$$|\mathbf{x}_u|^2 = E = 1 \quad \text{for all times.} \tag{7.1}$$

The coefficients F and G can vary, subject to $G - F^2 > 0$. Neighboring fibers can change their transverse spacing and their relative angle. Thus one family alone does not fix the surface metric.

Locally, many surface shapes admit coordinates with $E = 1$: choose a smooth nonintersecting family of curves and use arc length separately along each. This explains why one family imposes little purely local restriction on shape. It does not mean an arbitrary finite material sheet with fixed endpoints and boundary attachments can reach every shape. Global lengths, boundary conditions, and regularity can impose additional restrictions.

7.3 Two families fix lengths but permit shear

Use the two transverse families as coordinate curves $u =$ constant and $v =$ constant. Inextensibility gives

$$E(u, v, t) = E_0(u, v), \qquad G(u, v, t) = G_0(u, v), \tag{7.2}$$

while F can still vary. By (3.5), this remaining freedom is the angle between the two families. A woven net can shear by changing its crossing angles even when its threads do not stretch.

A special case is a Chebyshev net, with

$$E = G = 1, \quad F = \cos\phi, \qquad \mathrm{d}s^2 = \mathrm{d}u^2 + 2\cos\phi \mathrm{d}u\mathrm{d}v + \mathrm{d}v^2. \tag{7.3}$$

This normalization is a special property, not a coordinate choice available for every two-family network. Separate relabelings of u and v cannot generally remove arbitrary two-variable dependence from both E and G.

For $0 < \phi < \pi$, the area element of a Chebyshev net is

$$\mathrm{d}A = \sin\phi \mathrm{d}u\mathrm{d}v. \tag{7.4}$$

An initially square grid can shrink in area by shearing toward a narrow rhombus, even though both side lengths remain unchanged. At $\phi = 0$ or π, the coordinate directions become parallel and the surface parametrization degenerates.

Example 7.1 (A planar shearing net)**.** Set

$$\mathbf{x}(u, v, t) = \big(u + v\cos\phi(t),\ v\sin\phi(t),\ 0\big).$$

Then $E = G = 1$, $F = \cos\phi(t)$. Both fiber families are inextensible; their crossing angle and the area change. This gives a complete explicit motion realizing the remaining metric freedom.

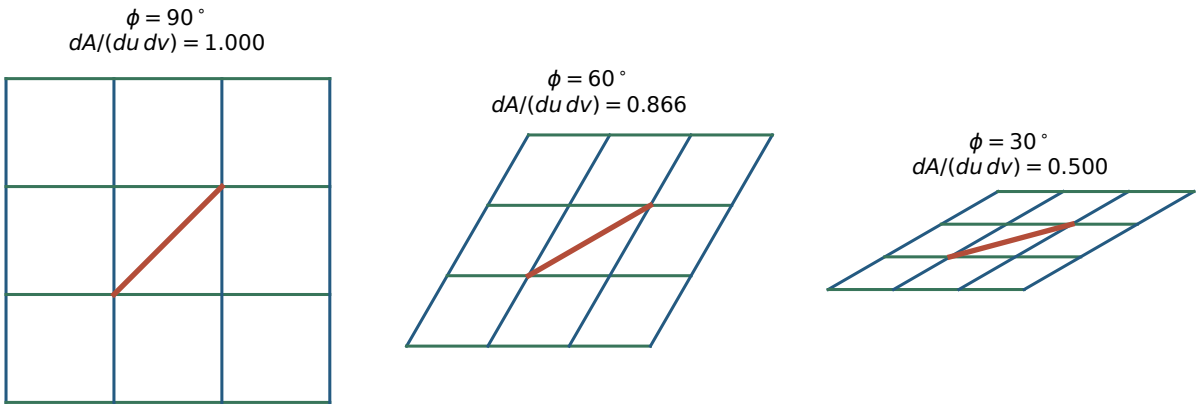


Figure 7.1: Two inextensible fiber families permit a change of crossing angle and area. The red diagonal changes length as the net shears; making that third material direction inextensible removes this freedom. Every blue and green unit segment retains length one.

7.4 How shear variation produces Gaussian curvature

For a spatially varying Chebyshev angle $\phi(u, v)$, curvature is determined by

$$\boxed{K = -\frac{\phi_{uv}}{\sin\phi}.} \tag{7.5}$$

Here is a short derivation. Define an orthonormal frame

$$\mathbf{e}_1 = \mathbf{x}_u, \qquad \mathbf{e}_2 = \frac{\mathbf{x}_v - \cos\phi\,\mathbf{x}_u}{\sin\phi}.$$

Since $E_v = G_u = 0$, the tangential part of $\mathbf{x}_{uv}$ is zero: its inner products with both $\mathbf{x}_u$ and $\mathbf{x}_v$ vanish. Thus $\omega(\partial_v) = 0$. Also

$$\langle \mathrm{D}_u\mathbf{x}_u, \mathbf{x}_v\rangle = \partial_u F = -\sin\phi\,\phi_u,$$

while $\mathrm{D}_u\mathbf{x}_u \perp \mathbf{x}_u$. Hence $\omega(\partial_u) = -\phi_u$, and

$$\omega = -\phi_u \mathrm{d}u, \qquad \mathrm{d}\omega = \phi_{uv}\mathrm{d}u \wedge \mathrm{d}v.$$

Use $\mathrm{d}\omega = -K\,\mathrm{d}A$ and $\mathrm{d}A = \sin\phi \mathrm{d}u\mathrm{d}v$ to obtain (7.5). A spatially constant shear angle gives zero Gaussian curvature. More generally, the crossing-angle field carries intrinsic geometric information.

7.5 A third independent family fixes the entire metric

Let a third family be described in the material grid by curves whose material tangent has components (a, b). If its length per unit material parameter is fixed at ℓ_0, then

$$Ea^2 + 2Fab + Gb^2 = \ell_0^2. \tag{7.6}$$

The first two families already determine E and G. Wherever $ab \neq 0$,

$$\boxed{F = \frac{\ell_0^2 - Ea^2 - Gb^2}{2ab}.} \tag{7.7}$$

If the third material parameter is its conserved arc length, $\ell_0 = 1$, which is the formulation in Peskin's notes. If $a = 0$ or $b = 0$, the third direction coincides with one of the existing directions at that point and gives no new information there.

The underlying linear algebra is simple. A symmetric 2×2 matrix has three independent entries. Squared lengths in three independent unoriented directions supply three independent linear measurements of those entries. The number of visible fiber families is not by itself sufficient; the directions must actually be distinct and mechanically inextensible at the point in question.

7.6 A fixed metric means isometry, not necessarily rigid motion

If E, F, G are fixed, the length integral of *every* material curve is fixed. Angles and areas are fixed as well. The motion is an isometric deformation of the surface. Because Gaussian curvature is intrinsic, it is preserved.

The extrinsic shape may still change. A flat sheet can roll into a cylinder without changing its metric. Thus preservation of all intrinsic lengths does not automatically force a patch to move as a rigid object in $\mathbb{R}^3$. Rigidity can follow in particular global settings with additional hypotheses, but it does not follow from the local three-family calculation alone.

If $\mathbf{V} = \mathbf{x}_t$ is the velocity field, differentiating $g_{ij} = \mathbf{x}_i \cdot \mathbf{x}_j$ gives

$$\partial_t g_{ij} = \partial_i \mathbf{V} \cdot \mathbf{x}_j + \mathbf{x}_i \cdot \partial_j \mathbf{V}. \tag{7.8}$$

An infinitesimal isometry satisfies that this expression vanishes. Decompose $\mathbf{V} = \mathbf{V}_T + V_n \mathbf{n}$. Since $\partial_i \mathbf{n} \cdot \mathbf{x}_j = -b_{ij}$, equation (7.8) becomes

$$\nabla_i (V_T)_j + \nabla_j (V_T)_i - 2V_n b_{ij} = 0. \tag{7.9}$$

Here $(V_T)_j = \mathbf{V}_T \cdot \mathbf{x}_j$, and $\nabla_i (V_T)_j = \partial_i (V_T)_j - \Gamma^k_{ij} (V_T)_k$. This equation makes the coupling explicit: normal motion of a curved sheet generally requires a compensating tangential velocity if its metric is to remain fixed.

Exercise 7.1. In the planar shearing net, compute the squared length of the diagonal material direction $(a, b) = (1, 1)$. Show directly why making that third family inextensible fixes ϕ on $0 < \phi < \pi$.

Chapter 8

From individual fibers to heart-wall equilibrium

8.1 A three-dimensional field of fiber directions

The heart wall occupies a volume, not just a surface. At each point let $\boldsymbol{\tau}(\mathbf{x})$ be a smooth unit vector representing the local fiber direction. Physically, a fiber direction is unoriented: $\boldsymbol{\tau}$ and $-\boldsymbol{\tau}$ describe the same alignment. For calculations we choose a consistent orientation locally. The stress constructed below is unchanged by reversing it.

A fiber path is an integral curve of this field:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}s} = \boldsymbol{\tau}(\mathbf{x}(s)). \tag{8.1}$$

Its parameter is arc length because $|\boldsymbol{\tau}| = 1$. For any scalar or Cartesian vector component $a(\mathbf{x})$,

$$\frac{\mathrm{d}a}{\mathrm{d}s} = (\boldsymbol{\tau} \cdot \nabla)a. \tag{8.2}$$

The operator $\boldsymbol{\tau} \cdot \nabla$ differentiates along the fibers. Applied to the direction field itself, it yields the curvature vector:

$$(\boldsymbol{\tau} \cdot \nabla)\boldsymbol{\tau} = \boldsymbol{\tau}_s.$$

Peskin's motivation is the varying orientation of myocardial fibers through the wall: near-circumferential directions in the middle and increasingly inclined directions toward the boundaries, with opposite signs of inclination on the two sides. The model seeks geometric consequences of fiber equilibrium rather than prescribing all pressure surfaces in advance.

8.2 Counting fibers through a cut

Let $\sigma(\mathbf{x})$ denote the number of fibers per unit area of a cut perpendicular to $\boldsymbol{\tau}$. If a small surface element has unit normal $\mathbf{n}$ and area $\mathrm{d}A$, its projected area perpendicular to the fibers is $|\boldsymbol{\tau} \cdot \mathbf{n}|\,\mathrm{d}A$. The number of fibers crossing it is

$$\sigma|\boldsymbol{\tau} \cdot \mathbf{n}|\,\mathrm{d}A.$$

Each fiber carries tension T. Accounting for the direction in which a fiber exits the region gives the traction vector

$$\mathbf{t}(\mathbf{n}) = \sigma T \boldsymbol{\tau}(\boldsymbol{\tau} \cdot \mathbf{n}). \tag{8.3}$$

This is a linear function of $\mathbf{n}$, so the corresponding Cauchy stress tensor is

$$\boxed{\boldsymbol{\Sigma}_f = \sigma T\, \boldsymbol{\tau} \otimes \boldsymbol{\tau}.} \tag{8.4}$$

A dyadic product acts as $(\mathbf{a} \otimes \mathbf{b})\mathbf{c} = \mathbf{a}(\mathbf{b} \cdot \mathbf{c})$. Thus the formula reproduces (8.3). Its eigenvalues are $\sigma T, 0, 0$: this idealization supports tension only in the fiber direction. A real tissue can have other stress contributions, but they are omitted in this model.

8.3 Fiber conservation

Consider a narrow tube formed by a bundle of fibers. Its side boundary is tangent to the fibers, so no fibers cross the sides. If fibers neither begin, end, nor branch within the tube, the number through every transverse section is the same. The local conservation law is

$$\boxed{\operatorname{div}(\sigma \boldsymbol{\tau}) = 0.} \tag{8.5}$$

It follows from zero net oriented fiber flux through every closed boundary and the divergence theorem. Expanding gives

$$\frac{\mathrm{d}\sigma}{\mathrm{d}s} + \sigma \operatorname{div} \boldsymbol{\tau} = 0.$$

Thus σ need not be constant along a fiber. If a bundle spreads apart, its transverse density decreases. Conservation concerns the product of density with the bundle's cross-sectional area.

The model is local where an orientation is chosen. Actual fiber ends or branching would add sources to the conservation equation; they are excluded from this derivation.

8.4 Divergence of fiber stress, component by component

The total fiber force on a volume V is

$$\mathbf{E}_f = \int_{\partial V} \boldsymbol{\Sigma}_f \mathbf{n}\, \mathrm{d}A.$$

For component i, use the divergence theorem and the product rule:

$$\begin{aligned} (\operatorname{div} \boldsymbol{\Sigma}_f)_i &= \partial_j(\sigma T \tau_i \tau_j) \\ &= T\tau_i \partial_j(\sigma \tau_j) + \sigma \tau_j \partial_j(T \tau_i) \\ &= \sigma \frac{\mathrm{d}}{\mathrm{d}s}(T \tau_i), \end{aligned} \tag{8.6}$$

where the last equality uses (8.5). Therefore the fiber force density is

$$\boxed{\mathbf{F}_f = \operatorname{div} \boldsymbol{\Sigma}_f = \sigma T_s \boldsymbol{\tau} + \sigma T \boldsymbol{\tau}_s.} \tag{8.7}$$

It has units force/volume: σ contributes inverse area, T contributes force, and $\boldsymbol{\tau}_s$ contributes inverse length. This is precisely the single-cable formula multiplied by the amount of fiber per transverse area.

8.5 Adding pressure and the free-sliding assumption

Include an isotropic compressive stress $-p\mathrm{Id}$, neglect inertia and body forces, and impose equilibrium:

$$\operatorname{div}(\boldsymbol{\Sigma}_f - p\mathrm{Id}) = 0, \qquad \sigma \frac{\mathrm{d}}{\mathrm{d}s}(T\boldsymbol{\tau}) = \nabla p. \tag{8.8}$$

Without further assumptions, dotting this equation with $\boldsymbol{\tau}$ gives

$$\sigma T_s = \frac{\mathrm{d}p}{\mathrm{d}s}. \tag{8.9}$$

Equilibrium alone therefore does not force pressure to be constant along a fiber.

The idealization used here adds the free-sliding, constant-tension condition $T_s = 0$. It is the continuum counterpart of a fiber with no longitudinal load transmitted by its constraints. Under this assumption,

$$\boxed{\sigma T \boldsymbol{\tau}_s = \nabla p.} \tag{8.10}$$

Taking a tangent dot product and using $\boldsymbol{\tau} \cdot \boldsymbol{\tau}_s = 0$ yields

$$\frac{\mathrm{d}p}{\mathrm{d}s} = 0. \tag{8.11}$$

Thus each fiber lies in a level set of the wall-pressure field.

8.6 Why the fibers are geodesics of pressure surfaces

Where $\nabla p \neq 0$, the implicit-function theorem says that $p = p_0$ is locally a smooth surface. Equation (8.10) says the fiber curvature vector is normal to it. Therefore the fiber is a geodesic on that pressure surface.

This answers a potentially confusing statement: a curved fiber inside a three-dimensional body is a geodesic *of a particular two-dimensional surface*, not a geodesic of ordinary Euclidean three-space. Euclidean geodesics are straight lines. The pressure foliation supplies the surfaces on which the intrinsic straightness occurs.

If the normal is chosen as $\mathbf{n}_p = \nabla p/|\nabla p|$, then

$$k_n = \frac{|\nabla p|}{\sigma T} > 0 \tag{8.12}$$

where $\sigma T > 0$. The principal normal aligns with increasing pressure. Choosing the opposite normal changes this signed normal curvature but not the physical curvature vector.

At a pressure critical point, $\nabla p = 0$, equation (8.10) gives $\boldsymbol{\tau}_s = 0$ if $\sigma T > 0$. The fiber is instantaneously straight, but pressure is not a valid local coordinate there, and the level-set geodesic description may cease to be regular.

8.7 A cylindrical wall as a complete consistency check

Let the fibers on each cylinder of radius r be helices of fixed angle on that cylinder. Write

$$\boldsymbol{\tau} = \cos\alpha(r)\,\widehat{\boldsymbol{\theta}} + \sin\alpha(r)\,\widehat{\mathbf{z}}.$$

Along a fiber, r and $\alpha(r)$ are constant. Since $\mathrm{d}\widehat{\boldsymbol{\theta}}/\mathrm{d}s = -\theta_s \widehat{\mathbf{r}}$ and $\theta_s = \cos\alpha/r$,

$$\boldsymbol{\tau}_s = -\frac{\cos^2\alpha}{r}\widehat{\mathbf{r}}.$$

For a radial pressure field, (8.10) becomes

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{\sigma T \cos^2\alpha}{r}. \tag{8.13}$$

If σT and α are constant, then

$$p(r) = p(r_0) - \sigma T \cos^2\alpha \log(r/r_0).$$

Pressure decreases outward, balancing the inward curvature force. Purely axial fibers have zero curvature and cannot support this radial pressure gradient by this mechanism alone.

The scalar p in these equations is a field in the wall. Relating it to ventricular cavity pressure requires boundary tractions and additional modeling information. A single pressure-surface calculation does not determine that relationship automatically.

Chapter 9

Coordinates adapted to fibers and pressure

9.1 Choosing coordinates that express the equilibrium geometry

Where pressure surfaces are regular and the fibers form a smooth local family, use coordinates

$$\mathbf{x} = \mathbf{x}(s, v, p).$$

Here s is arc length along a fiber; v labels fibers within one pressure surface; and p labels the pressure surfaces themselves. Along a fiber, v and p are constant, and

$$\mathbf{x}_s = \boldsymbol{\tau}, \qquad |\mathbf{x}_s| = 1. \tag{9.1}$$

This construction is local. To define it, choose a transverse initial surface, label its fiber intersections by (v, p), then follow the fiber flow for distance s. A global construction can fail if the chosen section misses fibers, if pressure has critical points, or if coordinates cease to be one-to-one.

The coordinate basis need not be orthogonal. In particular, $\mathbf{x}_p$ need not point in the direction of ∇p; it can contain a tangential part. Confusing a coordinate direction with a gradient is a common source of errors.

9.2 The Jacobian and the reciprocal basis

Define the signed Jacobian

$$J = \det[\mathbf{x}_s \ \mathbf{x}_v \ \mathbf{x}_p] = \mathbf{x}_s \cdot (\mathbf{x}_v \times \mathbf{x}_p). \tag{9.2}$$

Regularity means $J \neq 0$. A coordinate box has physical volume $\mathrm{d}V = |J|\mathrm{d}s\mathrm{d}v\mathrm{d}p$. The absolute value belongs to ordinary volume; the signed J belongs to oriented identities.

The scalar fields $s(\mathbf{x}), v(\mathbf{x}), p(\mathbf{x})$ are inverse coordinates. The chain rule gives

$$\nabla p \cdot \mathbf{x}_s = 0, \qquad \nabla p \cdot \mathbf{x}_v = 0, \qquad \nabla p \cdot \mathbf{x}_p = 1.$$

The cross product $\mathbf{x}_s \times \mathbf{x}_v$ is perpendicular to the first two basis vectors. Dividing it by its dot product with the third therefore constructs precisely the vector satisfying these three equations:

$$\boxed{\nabla p = \frac{\mathbf{x}_s \times \mathbf{x}_v}{J}.} \tag{9.3}$$

Similarly,

$$\nabla s = \frac{\mathbf{x}_v \times \mathbf{x}_p}{J}, \qquad \nabla v = \frac{\mathbf{x}_p \times \mathbf{x}_s}{J}. \tag{9.4}$$

These are the reciprocal-basis formulas. They require no orthogonality assumption; they are simply the inverse of a three-dimensional change-of-basis matrix.

Example 9.1 (Why a coordinate vector is not its reciprocal)**.** For the linear map $\mathbf{x}(s, v, p) = (s + p, v, p)$, $\mathbf{x}_p = (1, 0, 1)$, whereas $\nabla p = (0, 0, 1)$. The vectors differ, but their dot product is one, and ∇p is perpendicular to $\mathbf{x}_s = (1, 0, 0)$ and $\mathbf{x}_v = (0, 1, 0)$, exactly as required.

9.3 Conservation makes a coefficient constant along each fiber

A transverse surface at fixed s has oriented area vector $\mathbf{x}_v \times \mathbf{x}_p \mathrm{d}v\mathrm{d}p$. Its oriented fiber count is

$$\sigma\boldsymbol{\tau} \cdot (\mathbf{x}_v \times \mathbf{x}_p)\mathrm{d}v\mathrm{d}p = \sigma J \mathrm{d}v\mathrm{d}p.$$

Following the same tube to another value of s does not change this count. Thus

$$\partial_s(\sigma J) = 0. \tag{9.5}$$

Equivalently, the coordinate formula for divergence is $\operatorname{div}(\sigma\boldsymbol{\tau}) = J^{-1}\partial_s(J\sigma)$, because the contravariant components of $\boldsymbol{\tau} = \mathbf{x}_s$ are $(1, 0, 0)$. The sign of a nonzero J is constant on a connected regular chart, so this statement agrees with the physical conservation of the unsigned count $\sigma|J|$.

The assumed constant fiber tension gives $T_s = 0$. Hence

$$\partial_s(\sigma T J) = 0. \tag{9.6}$$

Define

$$\Lambda(v, p) = \frac{1}{\sigma T J}. \tag{9.7}$$

Substituting (9.3) into $\sigma T \boldsymbol{\tau}_s = \nabla p$ yields

$$\boldsymbol{\tau}_s = \Lambda(\boldsymbol{\tau} \times \mathbf{x}_v). \tag{9.8}$$

The pressure-gradient vector has disappeared from the formula, but its effect remains in the pressure-adapted coordinates and in Λ.

9.4 Fixing the cross-product convention once and for all

The axisymmetric calculation is particularly convenient with the cross product in the reverse order. Define

$$\lambda = -\Lambda = -\frac{1}{\sigma T J}. \tag{9.9}$$

Then the same physical equation is

$$\boxed{\mathbf{x}_s = \boldsymbol{\tau}, \qquad \boldsymbol{\tau}_s = \lambda(v, p)(\mathbf{x}_v \times \boldsymbol{\tau}).} \tag{9.10}$$

This is the convention used in the next chapter and in the final shape ODEs. If instead one defines the coefficient to be $+1/(\sigma T J)$, the cross product must be $\boldsymbol{\tau} \times \mathbf{x}_v$. One cannot reverse the cross product and retain the same signed coefficient.

This sign deserves attention when reading the original scan: the Jacobian identity on printed p. 143 has $\boldsymbol{\tau} \times \mathbf{x}_v$, while the subsequent axisymmetric equations use $\mathbf{x}_v \times \boldsymbol{\tau}$. The two symbols Λ and $\lambda = -\Lambda$ used here make that change of convention explicit and keep the downstream algebra consistent.

If v is an angle and hence dimensionless, then J has units length^2/pressure, $\sigma T J$ has units length^2, and λ has units length^{-2}. This will dictate the nondimensional parameter λr_0^2.

9.5 What is invariant under relabeling fibers?

Let $\widetilde{v} = f(v, p)$, with $f_v \neq 0$, and leave s, p unchanged. Then

$$\mathbf{x}_{\widetilde{v}} = \frac{\mathbf{x}_v}{f_v}, \qquad \widetilde{J} = \frac{J}{f_v}, \qquad \widetilde{\lambda} = f_v \lambda.$$

Thus $\widetilde{\lambda}(\mathbf{x}_{\widetilde{v}} \times \boldsymbol{\tau}) = \lambda(\mathbf{x}_v \times \boldsymbol{\tau})$. The scalar coefficient depends on the labeling convention; the physical vector equation does not. Fixing v to be the azimuth at the equatorial section removes this freedom in the axisymmetric calculation.

9.6 An immediate conserved quantity on each pressure surface

At fixed p, the metric in (s, v) has $E = |\mathbf{x}_s|^2 = 1$. Let $F = \mathbf{x}_s \cdot \mathbf{x}_v$. Then

$$\begin{aligned} F_s &= \mathbf{x}_{ss} \cdot \mathbf{x}_v + \mathbf{x}_s \cdot \mathbf{x}_{sv} \\ &= \lambda(\mathbf{x}_v \times \mathbf{x}_s) \cdot \mathbf{x}_v + \tfrac{1}{2}\partial_v |\mathbf{x}_s|^2 = 0. \end{aligned}$$

Therefore

$$\mathbf{x}_s \cdot \mathbf{x}_v \quad \text{is constant along each fiber.} \tag{9.11}$$

Geometrically this expresses the compatibility of a family of unit-speed geodesics with its transverse label. Under axial symmetry it becomes the angular-momentum, or Clairaut, integral $r^2 \theta_s = L$.

The equations govern one pressure surface at a time, but a wall requires a regular family of such surfaces. Neighboring surfaces must fit together with a nonzero Jacobian and suitable boundary tractions. Choosing an arbitrary shape independently at each pressure value does not guarantee a valid three-dimensional tissue configuration.

Chapter 10

The axisymmetric heart-wall shape equation

10.1 Cylindrical coordinates and their moving basis

Use cylindrical coordinates

$$\mathbf{x} = r\widehat{\mathbf{r}} + z\widehat{\mathbf{z}},$$

where

$$\widehat{\mathbf{r}} = (\cos\theta, \sin\theta, 0), \quad \widehat{\boldsymbol{\theta}} = (-\sin\theta, \cos\theta, 0), \quad \widehat{\mathbf{z}} = (0, 0, 1).$$

These vectors are orthonormal and right-handed: $\widehat{\mathbf{r}} \times \widehat{\boldsymbol{\theta}} = \widehat{\mathbf{z}}$, $\widehat{\boldsymbol{\theta}} \times \widehat{\mathbf{z}} = \widehat{\mathbf{r}}$. They vary with azimuth:

$$\frac{\mathrm{d}\widehat{\mathbf{r}}}{\mathrm{d}s} = \theta'\widehat{\boldsymbol{\theta}}, \qquad \frac{\mathrm{d}\widehat{\boldsymbol{\theta}}}{\mathrm{d}s} = -\theta'\widehat{\mathbf{r}}, \qquad \frac{\mathrm{d}\widehat{\mathbf{z}}}{\mathrm{d}s} = 0. \tag{10.1}$$

Primes in this chapter denote derivatives with respect to fiber arc length s. Differentiation gives

$$\boldsymbol{\tau} = r'\widehat{\mathbf{r}} + r\theta'\widehat{\boldsymbol{\theta}} + z'\widehat{\mathbf{z}}, \qquad r'^2 + r^2\theta'^2 + z'^2 = 1. \tag{10.2}$$

The quantity $r\theta'$, not θ', is the physical circumferential component of a unit tangent.

10.2 What axial symmetry means for the fiber family

Assume that rotating a pressure surface and its fiber field about the z-axis leaves them unchanged. Choose v to be the azimuth at an equatorial section $s = 0, z = 0$. Then a local representation is

$$\mathbf{x}(s, v, p) = \big(r(s,p)\cos(v + \psi(s,p)), r(s,p)\sin(v + \psi(s,p)), z(s,p)\big). \tag{10.3}$$

At fixed (s, p), varying v only rotates the point, so

$$\mathbf{x}_v = r\widehat{\boldsymbol{\theta}}. \tag{10.4}$$

The label v remains constant along a fiber even though its physical azimuth $\theta = v + \psi(s, p)$ changes. Symmetry also makes λ independent of v, leaving $\lambda = \lambda(p)$. On a single pressure surface it is a constant.

Using the convention (9.10),

$$\begin{aligned}\boldsymbol{\tau}_s &= \lambda(r\widehat{\boldsymbol{\theta}} \times \boldsymbol{\tau}) \\ &= \lambda r z' \widehat{\mathbf{r}} - \lambda r r' \widehat{\mathbf{z}}.\end{aligned} \tag{10.5}$$

Directly differentiating (10.2), including the moving-basis terms, gives

$$\boldsymbol{\tau}_s = (r'' - r\theta'^2)\widehat{\mathbf{r}} + (r\theta'' + 2r'\theta')\widehat{\boldsymbol{\theta}} + z''\widehat{\mathbf{z}}. \tag{10.6}$$

Comparing components yields

$$\boxed{\begin{aligned} r'' &= r\theta'^2 + \lambda r z', \\ \theta'' &= -\frac{2r'}{r}\theta', \\ z'' &= -\lambda r r'. \end{aligned}} \tag{10.7}$$

This system holds where $r > 0$. The term $r\theta'^2$ is the cylindrical-coordinate correction, and $\lambda r z'$ is the radial component of the curvature load.

10.3 Checking preservation of unit speed

Define $V^2 = r'^2 + r^2\theta'^2 + z'^2$. Differentiate and substitute:

$$\begin{aligned}(V^2)' &= 2r'r'' + 2rr'\theta'^2 + 2r^2\theta'\theta'' + 2z'z'' \\ &= 2r'(r\theta'^2 + \lambda r z') + 2rr'\theta'^2 - 4rr'\theta'^2 - 2\lambda r r' z' = 0.\end{aligned}$$

Thus unit initial speed is preserved exactly. This check is both a proof that s remains arc length and a practical numerical diagnostic.

10.4 Two first integrals

The axial equation integrates directly:

$$z'' = -\lambda r r' = -\frac{\lambda}{2}(r^2)',$$

so

$$\boxed{z' = C - \frac{\lambda}{2}r^2.} \tag{10.8}$$

The integration constant C is dimensionless because z' is dimensionless and λ has units length^{-2}.

Multiplying the azimuthal equation by r^2 gives

$$(r^2\theta')' = 2rr'\theta' + r^2\theta'' = 0,$$

hence

$$\boxed{r^2\theta' = L.} \tag{10.9}$$

This is Clairaut's relation, now obtained directly from the mechanical equations. It also follows from (9.11) because $\boldsymbol{\tau} \cdot \mathbf{x}_v = r^2\theta'$.

Combining the two integrals with unit speed gives

$$\boxed{r'^2 = 1 - \frac{L^2}{r^2} - \left(C - \frac{\lambda}{2}r^2\right)^2.} \tag{10.10}$$

The right side must be nonnegative wherever the fiber exists. The equation is often viewed as a one-dimensional energy equation, with an effective potential $L^2/r^2 + (C - \lambda r^2/2)^2$. Here "energy" means a mathematical first integral; it is not an additional claim about metabolic or elastic energy.

10.5 Why the substitution $\rho = r^2$ simplifies the algebra

Set $\rho = r^2$. Then

$$\rho' = 2rr', \qquad \frac{1}{2}\rho'' = rr'' + r'^2.$$

Use the radial equation and unit speed:

$$\begin{aligned}\frac{1}{2}\rho'' &= r^2\theta'^2 + \lambda r^2 z' + r'^2 \\ &= 1 - z'^2 + \lambda\rho z' \\ &= 1 - \left(C - \frac{\lambda}{2}\rho\right)^2 + \lambda\rho\left(C - \frac{\lambda}{2}\rho\right) \\ &= (1 - C^2) + 2C\lambda\rho - \frac{3}{4}\lambda^2\rho^2.\end{aligned}$$

Therefore

$$\boxed{\frac{1}{2}\rho'' = (1 - C^2) + 2C\lambda\rho - \frac{3}{4}\lambda^2\rho^2.} \tag{10.11}$$

Multiply by ρ'. The left side is the derivative of $\frac{1}{4}\rho'^2$. Integrating yields

$$\frac{1}{4}\rho'^2 = W(\rho) + a, \qquad W(\rho) = (1 - C^2)\rho + C\lambda\rho^2 - \frac{1}{4}\lambda^2\rho^3. \tag{10.12}$$

The same relation follows more transparently from (10.10):

$$\frac{1}{4}\rho'^2 = \rho r'^2 = \rho\left[1 - \left(C - \frac{\lambda}{2}\rho\right)^2\right] - L^2 = W(\rho) - L^2.$$

Thus the apparently new integration constant is

$$a = -L^2 \le 0. \tag{10.13}$$

Keeping this constraint prevents spurious scalar solutions that cannot correspond to real azimuthal motion of a unit-speed fiber.

10.6 Equatorial initial data and fiber angle

Suppose the chosen equatorial section is a radial turning point:

$$r(0) = r_0 > 0, \quad r'(0) = 0, \quad z(0) = 0.$$

Choose the fiber orientation so that the circumferential component is nonnegative, and let α_0 be the signed angle from the circumferential direction toward positive z. Then

$$z'(0) = \sin\alpha_0, \qquad r_0\theta'(0) = \cos\alpha_0. \tag{10.14}$$

It follows that

$$\rho_0 = r_0^2, \quad C = \sin\alpha_0 + \frac{\lambda}{2}\rho_0, \quad L = r_0\cos\alpha_0, \quad W(\rho_0) = L^2. \tag{10.15}$$

The radial first integral becomes

$$\rho' = \pm 2\sqrt{W(\rho) - W(\rho_0)}. \tag{10.16}$$

The sign is chosen separately on each interval of monotone radius. An initially decreasing radius uses the negative branch.

Equatorial data are a physical and geometric choice. Merely naming a section "equatorial" does not prove $r' = 0$; that condition is imposed here. Nor does $r' = 0$ by itself imply a maximum. Evaluate (10.7) initially:

$$r''(0) = \frac{\cos^2\alpha_0 + \lambda_0\sin\alpha_0}{r_0}, \qquad \lambda_0 = \lambda r_0^2. \tag{10.17}$$

A strict local equatorial maximum requires the numerator to be negative. Equality gives the cylindrical solution for these data, as shown below.

10.7 Eliminating arc length to obtain the profile equation

Where $\rho' \neq 0$, the chain rule gives

$$\boxed{\frac{\mathrm{d}z}{\mathrm{d}\rho} = \frac{C - \frac{\lambda}{2}\rho}{\pm 2\sqrt{W(\rho) - W(\rho_0)}}.} \tag{10.18}$$

A meridional curve $(r, z) = (\sqrt{\rho}, z(\rho))$, rotated about the axis, gives the pressure surface. The azimuthal fiber path can be reconstructed from

$$\frac{\mathrm{d}\theta}{\mathrm{d}\rho} = \frac{L}{\pm 2\rho\sqrt{W(\rho) - W(\rho_0)}}. \tag{10.19}$$

These integrals need not have elementary antiderivatives. Because their square root contains a cubic polynomial, they generally lead to elliptic integrals or numerical quadrature.

The denominator vanishes at a radial turning point. If the root is simple, the singularity of the profile quadrature is typically of inverse-square-root type and integrable. It means that ρ is a poor coordinate at the turning point, not that the arc-length-parametrized curve is necessarily singular. The original second-order ODE is the safer way to cross the turning point and select the next sign branch.

A first-order square-root equation initialized exactly at a root can also have misleading stationary continuations if one ignores the second-order equation. The vector ODE and its initial tangent resolve this ambiguity.

10.8 Scale versus shape: the dimensionless problem

Let

$$q = \frac{\rho}{r_0^2}, \qquad Z = \frac{z}{r_0}, \qquad \xi = \frac{s}{r_0}, \qquad \lambda_0 = \lambda r_0^2.$$

A dot below denotes $\mathrm{d}/\mathrm{d}\xi$. The complete reduced system is

$$\begin{aligned} C &= \sin\alpha_0 + \frac{1}{2}\lambda_0, \\ \frac{1}{4}\dot{q}^2 &= Q(q) := q\left[1 - \left(C - \frac{1}{2}\lambda_0 q\right)^2\right] - \cos^2\alpha_0, \\ \dot{Z} &= C - \frac{1}{2}\lambda_0 q, \qquad \dot{\theta} = \frac{\cos\alpha_0}{q}. \end{aligned} \tag{10.20}$$

At the equator, $q = 1$, $\dot{q} = 0$, and $Q(1) = 0$. The scale r_0 no longer appears. Thus, before fixing the initial fiber angle, there are two dimensionless shape parameters, λ_0 and α_0, equivalent to λ_0, C. Holding the angle fixed leaves a one-parameter family of shapes. Changing r_0 while replacing λ by λ_0/r_0^2 only rescales the solution.

Admissible profiles must remain in regions with $q > 0$ and $Q(q) \geq 0$. The polynomial's roots determine radial turning points. These algebraic restrictions are part of interpreting the formula, not optional details after solving it.

10.9 Two checks that reveal the meaning of the parameters

Example 10.1 (A cylinder)**.** For a constant-radius solution, $r' = r'' = 0$, $z' = \sin\alpha_0$, and $\theta' = \cos\alpha_0/r_0$. The radial equation requires

$$\lambda_0 = -\frac{\cos^2\alpha_0}{\sin\alpha_0}, \qquad \sin\alpha_0 \neq 0. \tag{10.21}$$

This makes (10.17) vanish. The fibers are precisely circular helices on a cylinder. A purely circumferential tangent has $\sin\alpha_0 = 0$; in that case $\mathbf{x}_s$ and $\mathbf{x}_v$ coincide in direction and the adapted (s, v) surface chart degenerates. The three-dimensional circular-fiber cylinder of Chapter 8 still makes sense, but this particular coordinate construction does not cover it.

Example 10.2 (The axis is inaccessible to nonmeridional geodesics)**.** From unit speed and $L = r_0\cos\alpha_0$,

$$\frac{L^2}{r^2} \leq 1 \quad \Longrightarrow \quad r \geq r_0|\cos\alpha_0|.$$

If $\cos\alpha_0 \neq 0$, the fiber and the pressure-surface patch generated by its rotations cannot continue all the way to $r = 0$ while this smooth, symmetric, single-family description remains valid. A radial turning point at positive radius is therefore expected. When $L = 0$, reaching the axis is no longer forbidden by this integral, but smooth closure still requires additional regularity checks at the axis.

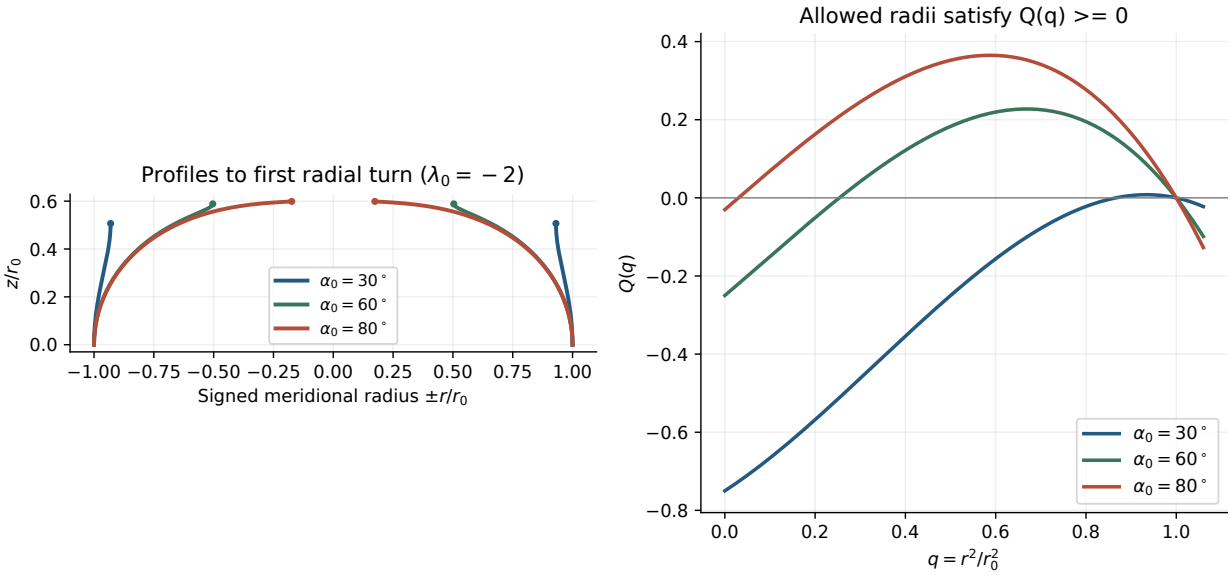


Figure 10.1: Dimensionless solutions for $\lambda_0 = -2$ and three equatorial fiber angles. Left: mirrored meridional profiles from the equator to the first radial minimum, marked by dots. They are partial pressure-surface profiles, not closed heart shapes. Right: the corresponding radial polynomial; only intervals with $Q \geq 0$ are admissible.

10.10 What remains to determine a whole heart wall

The calculation produces a family of candidate pressure-surface profiles, not a unique ventricle. A full wall would require the functions $r_0(p)$, $\alpha_0(p)$, and $\lambda_0(p)$, together with appropriate boundary tractions, compatibility between neighboring surfaces, and a treatment of regions where the idealized coordinates fail.

The relation $\lambda = -1/(\sigma T J)$ also ties the shape parameters to fiber density, tension, and inter-surface spacing. It is not enough to prescribe all those independently. Active contraction, passive material stresses, and dynamic motion would require further constitutive and kinematic equations.

The conceptual achievement is the chain of implications: fiber conservation and tension balance select pressure surfaces; fibers are geodesics on those surfaces; rotational symmetry supplies first integrals; and those integrals reduce the geometry to a one-dimensional profile equation. The remaining freedom is exactly where further biological or mechanical information must enter.

Chapter 11

Worked problems and reconstruction exercises

11.1 Resultant force on a circular cable

For Exercise 2.1, parametrize the arc by polar angle $\phi \in [-a, a]$:

$$\mathbf{x}(\phi) = R(\cos\phi, \sin\phi, 0), \qquad \mathrm{d}s = R\,\mathrm{d}\phi.$$

The curvature force per length is $-T(\cos\phi, \sin\phi, 0)/R$. Therefore

$$\begin{aligned}\mathbf{F} &= -T\int_{-a}^{a}(\cos\phi, \sin\phi, 0)\,\mathrm{d}\phi \\ &= (-2T\sin a, 0, 0).\end{aligned}$$

The unit tangent is $\boldsymbol{\tau} = (-\sin\phi, \cos\phi, 0)$, so $T\boldsymbol{\tau}(a) - T\boldsymbol{\tau}(-a) = (-2T\sin a, 0, 0)$, exactly the same result. The distributed force law is thus the infinitesimal form of the endpoint balance, not an independent assumption.

For Exercise 2.2, a helix has

$$\mathbf{f} = -\frac{TR}{R^2 + b^2}\widehat{\mathbf{r}}.$$

Since $\mathrm{d}s = \sqrt{R^2 + b^2}\,\mathrm{d}\phi$, its force per unit azimuth is

$$\frac{\mathrm{d}\mathbf{F}}{\mathrm{d}\phi} = -\frac{TR}{\sqrt{R^2 + b^2}}\widehat{\mathbf{r}}.$$

The first quantity has units force/length; the second has units force because radians are dimensionless. A parameter derivative is not automatically a derivative per unit physical length.

11.2 Classifying asymptotic directions

For Exercise 3.1, a unit direction in principal coordinates has normal curvature $k_n = k_1\cos^2\psi + k_2\sin^2\psi$. If $k_1k_2 > 0$, both terms have the same strict sign, so no asymptotic direction exists. If $k_1k_2 < 0$, solving gives

$$\tan^2\psi = -\frac{k_1}{k_2} > 0.$$

There are two distinct unoriented tangent lines. If exactly one principal curvature vanishes, only its principal direction is asymptotic. If both vanish, every direction is asymptotic at that point.

Existence of a direction at a point is not the same as constructing a complete curve. A smooth asymptotic direction field away from degeneracies can be integrated locally into curves, but global continuation can encounter singularities, boundaries, or changes in curvature type.

11.3 A latitude circle: all three curvatures and its holonomy

For Exercise 5.1, a latitude at colatitude ϑ has Euclidean radius $R\sin\vartheta$. Thus

$$\kappa = \frac{1}{R\sin\vartheta}, \qquad k_n = -\frac{1}{R}$$

with the outward sphere normal. By the orthogonal curvature decomposition,

$$|k_g| = \sqrt{\kappa^2 - k_n^2} = \frac{|\cot\vartheta|}{R}.$$

For increasing azimuth and $\mathbf{m} = \mathbf{n} \times \boldsymbol{\tau}$, the signed value is $k_g = \cot\vartheta/R$. Only the equator has $k_g = 0$.

We can also solve its parallel transport explicitly. Let $\mathbf{e}_\vartheta$ point toward increasing colatitude and $\mathbf{e}_\varphi$ toward increasing azimuth. Their connection one-form on the spherical coordinate patch is

$$\omega = \cos\vartheta \, \mathrm{d}\varphi.$$

Indeed, $\partial_\varphi \mathbf{e}_\vartheta = \cos\vartheta \, \mathbf{e}_\varphi$, while $\partial_\vartheta \mathbf{e}_\vartheta$ is normal. If a parallel vector makes angle θ with $\mathbf{e}_\vartheta$,

$$\frac{\mathrm{d}\theta}{\mathrm{d}\varphi} = -\cos\vartheta.$$

After one circuit, $\Delta\theta = -2\pi\cos\vartheta$, equivalent modulo 2π to

$$\beta = 2\pi(1 - \cos\vartheta).$$

The northern cap has area $2\pi R^2(1-\cos\vartheta)$, so this is exactly its integrated Gaussian curvature. The two real angle representatives differ by one full turn because the spherical coordinate frame is singular at the pole; the holonomy rotation itself is unambiguous.

11.4 The edge-stress estimate with numbers

For Exercise 6.1,

$$N \sim pR = 10, \qquad \sigma_{\text{interior}} \sim \frac{10}{0.1} = 100,$$

and

$$T_e \sim pRr = 20, \qquad \sigma_{\text{edge}} \sim \frac{20}{0.1^2} = 2000.$$

The stress ratio is $20 = r/h$. Increasing the effective edge area tenfold reduces its stress to 200 and the ratio to 2. These are scale calculations in the specified units; the geometric constants suppressed by $\sim$ are not determined.

11.5 Why a diagonal fiber removes the shear freedom

For Exercise 7.1, the diagonal tangent of the planar net is $\mathbf{x}_u + \mathbf{x}_v$. Its squared length is

$$|\mathbf{x}_u + \mathbf{x}_v|^2 = E + 2F + G = 2 + 2\cos\phi.$$

If this material diagonal is inextensible, its squared length remains fixed. Thus $\cos\phi$, and therefore $\phi \in (0, \pi)$, remains fixed. The two thread lengths alone did not determine the angle, but the diagonal length does. This is the metric version of the elementary fact that a triangle with all three side lengths fixed cannot shear.

11.6 A complete nondimensional shape calculation

Choose $r_0 = 1$, $\alpha_0 = 60^\circ$, and $\lambda_0 = -2$. Then

$$\sin\alpha_0 = \frac{\sqrt{3}}{2}, \quad \cos\alpha_0 = \frac{1}{2}, \quad C = \frac{\sqrt{3}}{2} - 1 \approx -0.1339745962.$$

The dimensionless radial polynomial is

$$Q(q) = q[1 - (q + C)^2] - \frac{1}{4}. \tag{11.1}$$

At $q = 1$, $q + C = \sqrt{3}/2$, so $Q(1) = 0$. The initial radial acceleration is

$$r''(0) = \frac{1}{4} - \sqrt{3} \approx -1.482050808,$$

which selects the initially decreasing-radius branch. Numerically, the next positive radial root is

$$q_{\min} \approx 0.2536314337, \qquad r_{\min} = \sqrt{q_{\min}} \approx 0.5036183413.$$

This lies above the Clairaut lower bound $r \geq 1/2$. The difference is caused by the nonzero axial component, which also uses part of the unit-speed budget.

The azimuth and height rates are

$$\dot\theta = \frac{1}{2q}, \qquad \dot Z = q + C.$$

On the admissible branch $q \in [q_{\min}, 1]$, the height rate stays positive. Integrating the original arc-length ODE reaches the first radial minimum at approximately

$$\xi \approx 1.250, \qquad Z \approx 0.589.$$

The displayed meridional curve stops there. It has not reached the axis, and it should not be labeled a closed ventricular surface.

11.7 How to reproduce and check the numerical calculation

The numerical implementation uses classical fourth-order Runge–Kutta. It is provided in `scripts/make_figures_and_check.py`, and evolves the state

$$y = (r, \theta, z, v_r, \omega, v_z),$$

with derivative

$$y' = \left(v_r, \omega, v_z, r\omega^2 + \lambda r v_z, -\frac{2v_r\omega}{r}, -\lambda r v_r \right).$$

For an arc-length step h, the four slopes are

$$k_1 = f(y), \qquad k_2 = f(y + hk_1/2),$$
$$k_3 = f(y + hk_2/2), \qquad k_4 = f(y + hk_3),$$

and the update is

$$y_{\text{new}} = y + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

No division by ρ' occurs, so the method can start at an equatorial turning point. Integration stops just after the first sign change of r'; the quoted turning height and arc length are correspondingly approximate.

Three diagnostics should remain constant:

$$I_1 = v_r^2 + r^2\omega^2 + v_z^2 = 1,$$
$$I_2 = v_z + \tfrac{1}{2}\lambda r^2 = C,$$
$$I_3 = r^2\omega = L.$$

For the three plotted examples at step $h = 10^{-3}$, the largest observed error in these invariants is below 10^{-8}. Halving the step for the 60° example changes the state at shared sample points by less than 10^{-9}. These checks test the implementation of the idealized equations; they do not validate the biological assumptions of the model.

11.8 Reconstruct the argument without looking at the formulas

The following problems are intended for active study. Each asks for a derivation rather than a recalled definition.

1. Starting from endpoint forces on a small cable segment, derive the distributed force and explain which sign corresponds to the force on the cable.
2. Starting from $\mathbf{w} \cdot \mathbf{n} = 0$ and the requirement that $\mathbf{w}_s$ be normal, derive parallel transport and prove preservation of inner products.
3. Starting from a surface parametrization, derive its metric, area element, second fundamental form, and normal-curvature quotient.
4. Use the normal component of edge-cable balance to prove the asymptotic-line condition. State exactly where positive cable tension and nonzero edge curvature enter the conclusions.
5. Count fibers across an oblique cut and derive the tensor $\sigma T \boldsymbol{\tau} \otimes \boldsymbol{\tau}$. Use conservation to simplify its divergence.
6. Derive $\nabla p = (\mathbf{x}_s \times \mathbf{x}_v)/J$ by testing dot products against the three coordinate basis vectors.
7. Recompute the cylindrical moving-basis derivatives and recover all three equations in (10.7).
8. Recover both first integrals, the ρ equation, and the constraint $a = -L^2$. Then identify which parameters change size and which change dimensionless shape.

A successful reconstruction should make clear where geometry ends and an assumption about force, material behavior, symmetry, or boundary data enters.

Appendix A

Notation, units, and a guide to the original chapter

A.1 Geometric notation

Symbol	Meaning
$\mathbf{x}$	Position or surface parametrization in Euclidean space.
s, u	Arc length when explicitly specified; u is also used as a general surface coordinate.
$\boldsymbol{\tau}$	Unit tangent to a fiber or curve.
$\kappa, \mathbf{n}_c$	Nonnegative space-curve curvature and principal normal.
τ_{F}	Torsion of a space curve.
$\mathbf{n}$	A chosen surface unit normal.
$\mathbf{m}$	Tangent conormal $\mathbf{n} \times \boldsymbol{\tau}$ along a curve.
E, F, G	Metric coefficients $\mathbf{x}_u^2, \mathbf{x}_u \cdot \mathbf{x}_v, \mathbf{x}_v^2$.
g_{ij}, g^{ij}	Metric matrix and its inverse.
b_{ij}, II	Coefficients and quadratic form of the second fundamental form.
$S = -\mathrm{d}N$	Shape operator; $N(\mathbf{x}) = \mathbf{n}(\mathbf{x})$ is the Gauss map.
k_1, k_2	Principal curvatures, the eigenvalues of S.
k_n, k_g	Signed normal and geodesic curvature of a surface curve.
K, H	Gaussian curvature $k_1 k_2$ and mean curvature $(k_1 + k_2)/2$.
Γ_{ij}^k	Christoffel symbols of the metric connection.
$\mathrm{D}/\mathrm{d}s$	Tangentially projected derivative along a surface curve.
ω	Connection one-form in a chosen oriented orthonormal frame; in the numerical state it is separately named angular speed.
β	Holonomy rotation, defined modulo 2π.

A.2 Mechanical quantities and dimensions

Let L denote a length unit and F a force unit; the integration constant L below is a different object.

Symbol	Dimensions	Meaning
T, T_e	F	Tension per fiber or in an edge cable.
$\mathcal{T}$	$\mathsf{F}/[v]$	Membrane fiber-strip tension per unit label.
N, N_f	F/L	Membrane stress resultant.
p	F/L^2	Pressure difference on a sheet or wall pressure field, as specified.
σ	L^{-2}	Fiber number per transverse area, treating fiber count as dimensionless.
σT	F/L^2	Fiber stress magnitude.
$\mathbf{F}_f$	F/L^3	Fiber force per tissue volume.
κ, k_n, k_g	L^{-1}	Curve curvatures.
K	L^{-2}	Gaussian curvature.
J	$\mathsf{L}^2/[p]$	Signed Jacobian for (s, v, p), with dimensionless v.
λ	L^{-2}	Coefficient for $\mathbf{x}_v \times \boldsymbol{\tau}$.
C	1	Axial first-integral constant.
L	L	Clairaut constant $r^2\theta'$.
ρ, W, a	L^2	Squared radius, radial cubic, and its additive integration constant.
λ_0, q, Z, ξ	1	Dimensionless shape coefficient, squared radius, height, and arc length.

A.3 A compact map of the main equations

The central geometric identities are

$$\begin{aligned}
\boldsymbol{\tau}_s &= \kappa \mathbf{n}_c = k_g \mathbf{m} + k_n \mathbf{n}, \\
\mathrm{d}s^2 &= E\mathrm{d}u^2 + 2F\mathrm{d}u\mathrm{d}v + G\mathrm{d}v^2, \\
\mathbf{w}_s &= -(\mathbf{w}\cdot\mathbf{n}_s)\mathbf{n} \quad \text{for parallel transport}, \\
K &= \det S, \qquad \beta = \int_D K \,\mathrm{d}A \pmod{2\pi}.
\end{aligned}$$

The mechanical-to-geometric chain is

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}s}(T\boldsymbol{\tau}) + \mathbf{q} &= 0 && \text{cable balance}, \\
\partial_u(\mathcal{T}\boldsymbol{\tau}) + p(\boldsymbol{\tau}\times\mathbf{x}_v) &= 0 && \text{one-family leaflet}, \\
\operatorname{div}(\sigma\boldsymbol{\tau}) &= 0 && \text{fiber conservation}, \\
\sigma T \boldsymbol{\tau}_s &= \nabla p && \text{constant-tension wall}, \\
\nabla p &= \frac{\mathbf{x}_s \times \mathbf{x}_v}{J} && \text{pressure coordinates}, \\
\boldsymbol{\tau}_s &= \lambda(\mathbf{x}_v \times \boldsymbol{\tau}) && \lambda = -1/(\sigma T J).
\end{aligned}$$

The final profile equation follows from these identities plus axial symmetry and the equatorial initial data.

A.4 Correspondence with the supplied scan

Printed page numbers in the source equal PDF page numbers plus 126.

Source pages	Topic	Developed here
127–128	Space curves, curvature, cable tension	Chapters 1–2
128–130	Spherical image and parallel transport	Chapters 3–4
130–133	Holonomy, additivity, Gaussian curvature	Chapter 4
133–134	Geodesics and freely sliding fibers	Chapter 5
134–136	Closure line, tangent leaflets, edge stress	Chapter 6
136–137	Pressure-loaded fiber-family equations	Chapter 6
137–138	One, two, and three inextensible families	Chapter 7
138–141	Fiber architecture, stress, continuum balance	Chapter 8
141–143	Pressure surfaces and adapted coordinates	Chapters 8–9
143–145	Cylindrical basis, symmetry, first integrals	Chapter 10
145–147	Squared-radius reduction, scaling, free parameters	Chapters 10–11

A.5 Notation changes that prevent collisions

The source writes k for space-curve curvature; here it is κ. Its principal normal $\mathbf{n}_1$ is $\mathbf{n}_c$. The cable, strip, and individual-fiber tensions are separated as $T_e, \mathcal{T}, T$ where useful. In the final calculation, the source's integration constant K is written C, leaving K exclusively for Gaussian curvature. Its polynomial $F(\rho)$ is written $W(\rho)$, leaving F available for the mixed metric coefficient. The equatorial fiber angle is α_0, so θ can consistently denote cylindrical azimuth.

The sign convention for the two cross-product forms of the pressure-coordinate equation is explained in Chapter 9. With that convention, all component equations and first integrals in Chapter 10 use the same signed λ.

Bibliography

[1] Charles S. Peskin. *Mathematical Aspects of Heart Physiology*. Courant Institute Lecture Notes, 1975. Chapter III, "The Differential Geometry of the Heart," pp. 127–147. Supplied scan: `CLN-Peskin1975-3.pdf`. Collection: https://math.nyu.edu/~peskin/heartnotes/index.html.